PDC-01: Control objectives, feedback and feedforward

Overview

Assumed: steady material and energy balances.

Control keeps a process near a desired operating condition while its inputs change. A heated, well-mixed tank shows the difference between rejecting a disturbance and following a new target, and between correcting an observed error and anticipating its cause.

Intuition

A steady design tells us how much heat is needed at one condition. It does not make that heat supply adjust when the feed becomes colder. Feedback watches the result; feedforward watches a disturbance and predicts the required correction.

The **controlled variable** is the quantity to regulate, the **manipulated variable** is an input we can adjust, and a **disturbance** is an input change outside this controller's authority. A **set point** is the desired controlled-variable value.

Follow a cold-feed disturbance through the tank. Colder liquid first reduces the energy arriving each second; stored energy then falls, so the tank cools. Feedback detects that temperature fall and raises heating. Feedforward measures the colder feed and raises heating before the tank needs to cool. This causal sequence explains why feedback needs a changed measurement and why feedforward needs a model of the disturbance's effect.

Derivation

Step 1: identify the physical balance. Let liquid density be ρ, specific heat capacity c_p, tank volume V, equal inlet/outlet volumetric flow q, feed temperature T_0, tank/outlet temperature T, and heating rate Q. Assume perfect mixing, constant properties and volume, no reaction, heat loss or appreciable mixer work. Energy accumulation gives

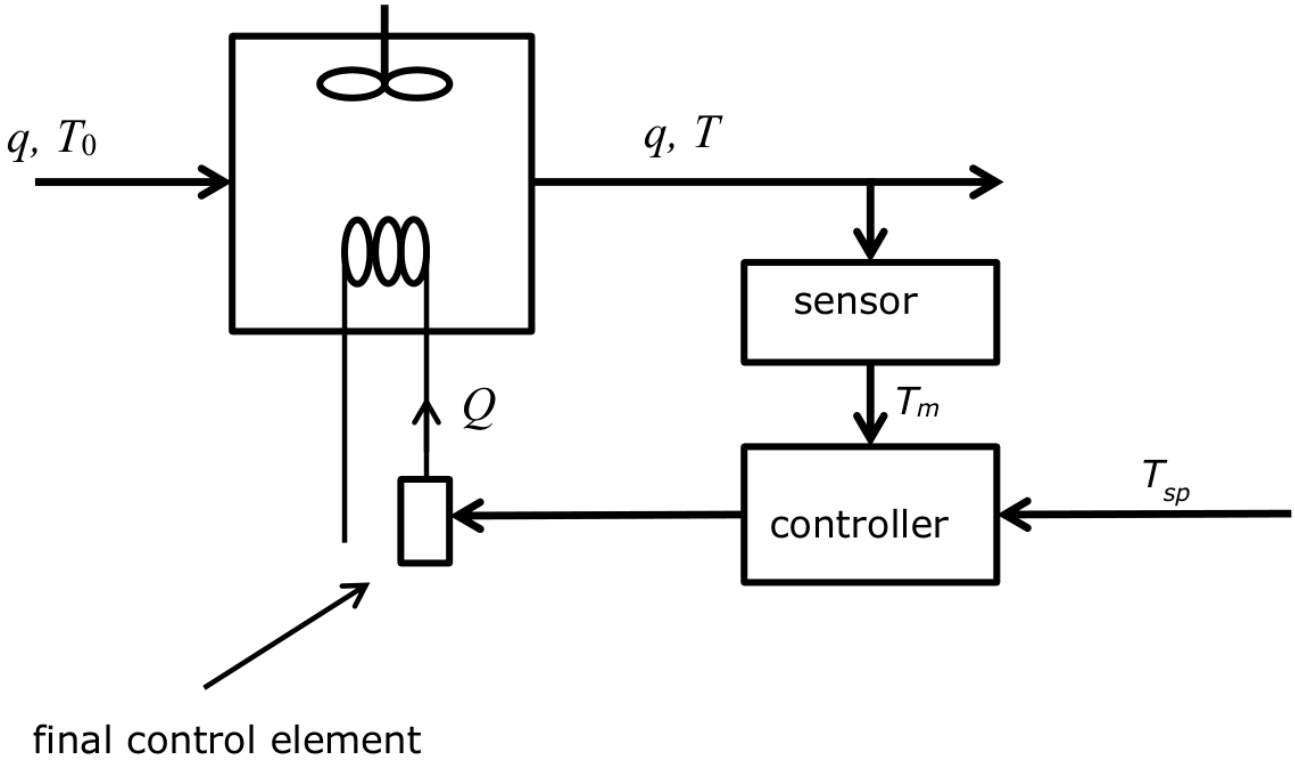
$$\rho V c_p \frac{dT}{dt} = \rho q c_p (T_0 - T) + Q. \tag{1}$$

Reason: inflowing liquid brings sensible heat, the outlet carries heat at the tank temperature, and the heater adds energy. At steady target temperature T_{sp}, set the derivative to zero:

$$Q_s = \rho q c_p (T_{sp} - T_{0s}). \tag{2}$$

Here subscript s denotes a steady operating value. This is the heat requirement, not a controller algorithm.

Step 2: define feedback. A sensor reports measured temperature T_m; the controller forms error $e = T_{sp} - T_m$ and commands the final control element, which changes Q. If T_m is too low, more heating should raise T; if it is too high, heating should fall.



Reason: the measurement closes a path from process output back to process input. Sensor and actuator dynamics are part of that path.

Step 3: define feedforward. If T_0 is measured, an ideal heat command for this simplified tank is

$$Q_{ff} = \rho q c_p (T_{sp} - T_0). \tag{3}$$

Reason: replacing the unknown disturbance by its measured value cancels its contribution to the energy balance. With this exact model and instantaneous heating, $\rho V c_p \dot{T} = \rho q c_p (T_{sp} - T)$, so a tank initially at the fixed target stays there.

Motivation and explanation

Regulator control holds the target fixed while rejecting a disturbance. **Servo control** changes the output to follow a changed target. The same loop can perform both; they are different tests of its behaviour.

An on/off thermostat switches heating on below its switching range and off above it. Heat storage and switching can sustain a temperature cycle even with constant outside conditions; feedback alone

does not guarantee a smooth response.

Feedback can respond to unmeasured disturbances but requires an error or measurement change before acting. Feedforward can act before an output error, but only for measured disturbances and only as well as the model and actuator permit. Combining them lets feedback correct residual modelling error.

Control objectives include safety constraints, operability and profitability. Inventory, pressure and temperature limits support operation; product quality and energy use support economics. Ordinary regulatory control is only one protective layer. Heater capacity, valve limits and process design constrain what any controller can achieve.

Worked example

Take water with $\rho = 1000$ kg m^{-3}, $c_p = 4200$ J kg^{-1} K^{-1} and $q = 0.010$ m^3 s^{-1}. Holding $T_{sp} = 50$ °C with $T_{0s} = 20$ °C requires $Q_s = 1.260$ MW. If the inlet falls to 15 °C, the model requires 1.470 MW, an increase of 0.210 MW.

A regulator makes that correction while keeping the target at 50 °C. A servo instead reacts to an operator changing the target. A real feedforward command would also need accurate flow and properties, sufficient heat capacity and allowance for sensor/actuator delay.

The extra heating has a simple meaning: the heat-capacity flow rate is $\rho q c_p = 42000$ W K^{-1}, so every kelvin lost from the feed needs that much extra power. The tank's stored heat determines how quickly an uncompensated error develops; it does not change the final power required. The ideal feedforward law preserves an existing fixed target, but an instantaneous target change cannot make stored liquid jump to a new temperature.

Link to other topics

The same accumulation balance will give the tank time constant. Feedback design then depends on the combined dynamics of tank, sensor and final control element.

Conclusion

Choose the objective and available measurements before choosing a controller. A correct steady heat requirement is the starting point for a dynamic control model.

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PDC-02: Choosing variables and interpreting distillation loops

Overview

Assumed: distillation equipment and the roles of reflux, reboiling and condensation.

Control-variable labels depend on the chosen process boundary and available actuators. A distillation column makes this concrete: pressure and inventories must be maintained while product compositions are regulated.

Intuition

A vessel level is an inventory signal: it tells us whether material is accumulating. A tray temperature can be a composition signal: at a known pressure and suitable mixture conditions, a changed composition changes the boiling temperature. These measurements solve different problems even though their loops interact.

To choose an actuator, imagine moving its valve and follow what happens to the measured quantity. Opening distillate withdrawal removes liquid from the drum directly. Increasing reflux also removes drum liquid, but sends it back into the column and changes separation. Reserving different actuators for inventory and quality avoids asking one valve to satisfy two independent objectives at once; it does not eliminate interactions.

Derivation

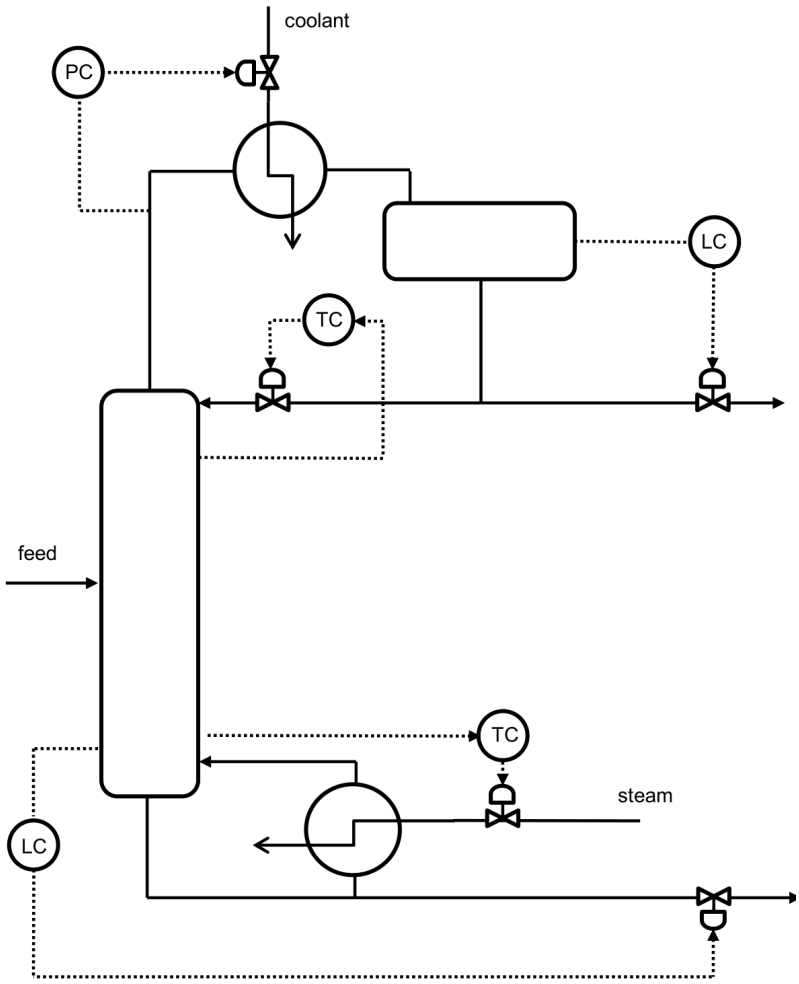
Step 1: specify boundaries. Include the column, condenser, reflux drum and reboiler in the process boundary. The feed is an external input; reflux is an internal circulation but remains an actuator-driven input to the column subsystem.

Step 2: connect inventory to a balance. If M_D is drum liquid mass and $\dot{m}_C$, $\dot{m}_R$, $\dot{m}_D$ are condensate, reflux and distillate mass flow rates, then

$$\frac{dM_D}{dt} = \dot{m}_C - \dot{m}_R - \dot{m}_D. \tag{4}$$

Reason: drum level can remain steady only when inflow matches the two outflows. Adjusting distillate flow changes accumulation directly. An analogous base balance connects bottom level to bottoms withdrawal.

Step 3: read the five loops. PC, LC and TC mean pressure, level and temperature controller. Dotted lines carry measurement/control information; solid lines carry material.



Controlled measurement	Manipulated input in this arrangement	Physical reason
Column overhead pressure	Condenser coolant flow	Heat removal changes condensation and vapour inventory
Reflux-drum level	Distillate withdrawal	Removes accumulated condensate
Upper-column temperature	Reflux flow	Changes internal liquid flow and upper separation
Lower-column temperature	Reboiler steam flow	Changes boil-up and lower separation
Column-base level	Bottoms withdrawal	Removes accumulated bottom liquid

Reason: each information path terminates at a valve that can influence its measurement. This is one possible pairing, not a universal rule.

At the old steady state, the drum inflow and outflows balance. If condensate inflow rises while both withdrawals stay fixed, the positive imbalance integrates into increasing mass and level. With no level-dependent outflow in this simplified balance, a higher level alone does not remove that imbalance. The level controller supplies the restoring action by changing withdrawal. A temperature

loop instead changes internal contacting and vapour–liquid equilibrium, so its physical pathway is different.

Motivation and explanation

The desired variables may be top and bottom composition, while the measured variables are tray temperatures. Temperature is an **inferential measurement** of composition; it must be justified for the mixture and pressure. Pressure changes, multicomponent feeds or weak temperature sensitivity can undermine the inference.

Feed flow, composition and temperature often act as disturbances because an upstream unit determines them. They become manipulated variables only if this controller is allowed to adjust them. Reflux changes both inventories and separation, and steam changes several tray conditions: five loops do not imply five independent processes.

Worked example

Suppose the reflux-drum level rises while its set point is unchanged. Increasing distillate withdrawal makes the right side of the drum balance smaller, opposing accumulation. Reducing reflux could also lower the level, but would disturb upper-column separation; the pairing above reserves reflux for the temperature loop.

If pressure rises, the pressure loop can increase condenser cooling within its available range. A temperature loop should not interpret every simultaneous temperature change as composition error: boiling temperature also depends on pressure.

Link to other topics

Phase equilibrium explains why pressure and composition affect tray temperature. Dynamic inventory balances explain why level loops are essential even when product specifications are the main economic objective.

Conclusion

A control structure is a set of physically justified measurement–actuator pairings. Its adequacy depends on interactions and dynamics as well as steady separation performance.

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PDC-03: CSTR variable identification and dynamic model closure

Overview

Assumed: well-mixed reactor mass and energy balances.

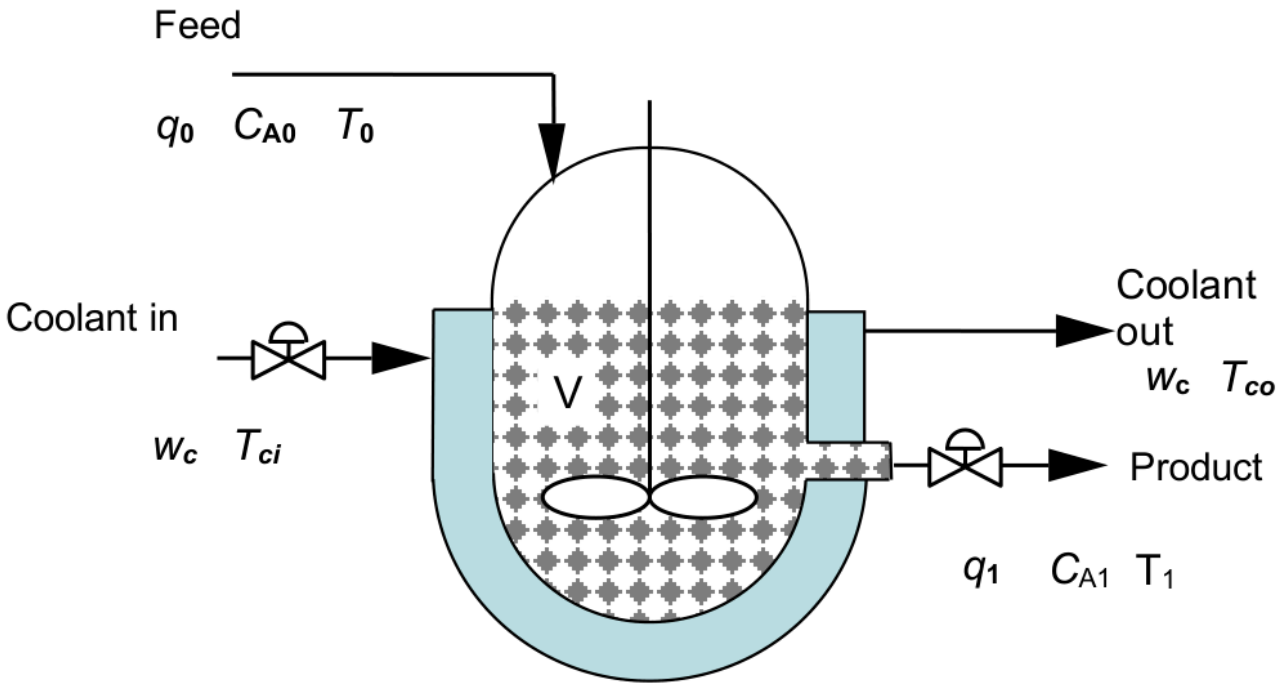
For a jacketed CSTR carrying out $A \rightarrow B$, identify inputs, outputs and states before simplifying its balances. Measurements and manipulability are engineering choices, rather than intrinsic properties of a variable.

Intuition

The reactor stores material and energy. Feed and coolant conditions drive its behaviour; concentration, temperature and volume describe its evolving contents. A product-flow valve is physically on an outlet line but may be treated as a model input because its position changes reactor accumulation.

Derivation

Step 1: label the equipment. Let q_0, q_1 be feed/product volumetric flow rates, c_{A0}, c_{A1} their concentrations of A, T_0, T_1 their temperatures, V reactor volume, w_c coolant mass flow, and T_{ci}, T_{co} coolant inlet/outlet temperatures.



Step 2: state one plausible control assignment. Let the upstream process determine q_0, c_{A0}, T_0. Use product withdrawal to control volume and coolant flow to control reactor temperature.

Variable	Model role under this assignment	Typical measurement and use
q_0	Disturbance input	Flowmeter; monitor or use for compensation
c_{A0}	Disturbance input	Online analyser or sampled analysis; may be unmeasured online
T_0	Disturbance input	Temperature sensor
q_1	Manipulated input	Flowmeter and product valve/pump
c_{A1}	Output/state	Analyser or sampled analysis; possible quality target
T_1	Controlled output/state	Temperature sensor; coolant-flow loop
V	Controlled output/state	Level measurement with geometry conversion
w_c	Manipulated input	Coolant flowmeter and valve
T_{ci}	Disturbance input	Temperature sensor; utility condition
T_{co}	Output of coolant subsystem	Temperature sensor; heat-duty monitoring

Reason: disturbances enter from outside the selected authority, whereas actuators can be commanded. An input can be measured; an output can be unmeasured. If q_1 instead follows a

gravity-outflow law, it becomes a dependent output and a different actuator must regulate volume.

Step 3: write the balances. For constant equal liquid densities, no volume change on mixing, and positive reaction-consumption rate $r(c_{A1}, T_1)$ per reactor volume,

$$\dot{V} = q_0 - q_1, \qquad \frac{d(V c_{A1})}{dt} = q_0 c_{A0} - q_1 c_{A1} - V r. \tag{5}$$

Reason: use the extensive amount $V c_{A1}$ while volume can change. Expand its derivative and substitute the volume balance:

$$V \dot{c}_{A1} = q_0 (c_{A0} - c_{A1}) - V r. \tag{6}$$

Reason: well-mixed withdrawal removes solute and solvent proportionally, so its direct concentration contribution cancels.

For constant liquid properties, negligible pressure work and mixer heating, heat of reaction ΔH_r per mole reacted, and positive cooling duty Q_c,

$$\rho V c_p \dot{T}_1 = \rho q_0 c_p (T_0 - T_1) + (-\Delta H_r) V r - Q_c. \tag{7}$$

Reason: the volume derivative cancels the outlet sensible-energy term in the same way. Exothermic reaction has $\Delta H_r < 0$ and therefore adds heat. Close the model with a rate law, geometry, valve/flow relations and a heat-transfer model. For example, $Q_c = U A_h (T_1 - T_c)$ uses heat-transfer coefficient U, area A_h and a representative jacket temperature T_c; a dynamic jacket needs its own energy balance.

The outlet-flow cancellation is physical, not just convenient algebra. Removing a small portion of perfectly mixed liquid removes the same fraction of A as of total volume, leaving the remaining concentration unchanged at that instant. It still changes V, so subsequent feed dilution and reaction per stored amount can evolve differently. A concentration is a ratio; the amount balance and volume balance must agree before an equation for that ratio is trustworthy.

Model closure means being able to calculate every right-hand side from the current states and specified inputs. Knowing V, c_{A1}, T_1 is insufficient if r or Q_c is still unknown. A rate law supplies r; a heat-transfer relation and, if needed, a jacket energy balance supply Q_c. Each additional stored quantity needs its own initial condition and balance.

Motivation and explanation

Steady state sets all accumulation terms to zero. It does not remove the need for the rate and heat-transfer relations. A temperature-dependent rate can make concentration and temperature strongly coupled; assuming constant k deliberately excludes that coupling.

The general control-volume energy balance stores internal energy and uses stream enthalpy, heat and work. If full chemical enthalpies include reaction energetics, do not also add a separate reaction-

heat term. Here sensible energy and explicit ΔH_r are a consistent alternative.

Worked example

A colder coolant inlet is a disturbance even when it is measured. It changes heat removal, which changes T_1, which may change reaction rate and c_{A1}. A coolant-flow adjustment is a manipulated input because the valve can be commanded.

To close a model of T_{co}, one cannot merely list that temperature as an independent input. Either apply a quasi-steady coolant balance $Q_c = w_c c_{pc}(T_{co} - T_{ci})$, with coolant heat capacity c_{pc}, or include jacket storage and its temperature distribution.

Link to other topics

Reactor modelling provides the balances; process control decides which inputs are available and which outputs matter. Transfer functions will describe the response of those outputs near an operating point.

Conclusion

A useful model begins with a boundary, a variable assignment and enough constitutive relations to determine the states. The labels alone do not close the dynamics.

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PDC-04: Mixing tanks and reacting first-order lags

Overview

Assumed: an isothermal CSTR balance and first-order reaction kinetics.

A well-mixed tank converts inlet-composition changes into a first-order lag. Adding reaction changes both the steady gain and the time constant; the hydrodynamic residence time remains V/q.

Intuition

Mixing spreads the arrival of new feed through existing contents. A change is felt immediately, but the old contents are only gradually displaced. Reaction removes A as well as flow removing it, so a concentration deviation disappears faster and a feed increase produces a smaller final concentration increase.

Derivation

Step 1: balance inert solute. With fixed volume V, flow q, inlet concentration c_0 and uniform tank/outlet concentration c, assume constant density, no reaction and perfect mixing:

$$V\dot{c} = q(c_0 - c), \qquad \tau\dot{c} + c = c_0, \quad \tau = V/q. \tag{8}$$

Reason: fixed volume gives $d(Vc)/dt = V\dot{c}$. Dividing by q makes the coefficient of c equal to one. In the standard first-order form $\tau\dot{c} + c = K_p c_0$, the derivative coefficient is the time constant and the right-side coefficient is the gain. Setting $\dot{c} = 0$ gives $c = c_0$, so an inert tank has unit gain.

The ratio V/q is also the mean residence time: stored liquid volume divided by volume replaced each second. Perfect mixing allows some new feed to leave quickly and some to remain longer, so it is an average, not an identical transit time for every fluid element.

Step 2: solve a feed step. Initially $c = c_s = c_{0s}$. For $t \geq 0$, the inlet changes permanently by Δc_0. Set $c' = c - c_s$; then $\tau\dot{c}' + c' = \Delta c_0$ with $c'(0) = 0$. Subtracting the old steady balance removes constant terms; because the tank starts there, its initial deviation is zero. First divide the ODE by τ, then multiply by $e^{t/\tau}$:

$$\frac{d}{dt}(e^{t/\tau}c') = \frac{\Delta c_0}{\tau}e^{t/\tau}.$$

Reason: the product rule gives $d(e^{t/\tau}c')/dt = e^{t/\tau}\dot{c}' + e^{t/\tau}c'/\tau$, exactly the left side after multiplication. The integrating factor is chosen to create that exact derivative. Integration from zero to t gives $e^{t/\tau}c' = \Delta c_0(e^{t/\tau} - 1)$, hence

$$c' = \Delta c_0(1 - e^{-t/\tau}). \tag{9}$$

Step 3: include first-order consumption. For constant rate constant $k > 0$, add loss kVc:

$$V\dot{c} = q(c_0 - c) - kVc, \qquad \frac{\tau}{1 + k\tau}\dot{c} + c = \frac{c_0}{1 + k\tau}. \tag{10}$$

Reason: reaction consumes kc per unit volume, hence kVc in the whole tank. Collecting terms gives $V\dot{c} + (q + kV)c = qc_0$. Divide by the coefficient of c, namely $q + kV$, to put this in standard form:

$$\tau_r = \frac{V}{q + kV} = \frac{\tau}{1 + k\tau}, \qquad K_p = \frac{q}{q + kV} = \frac{1}{1 + k\tau}.$$

The same total removal coefficient appears below both ratios. Storage divided by removal sets the response time; input coefficient divided by removal sets the settled sensitivity. These are different properties even though reaction reduces both here.

Step 4: transform the perturbation balance. The reacting equilibrium is $c_s = c_{0s}/(1 + k\tau)$, so it differs from the inert equilibrium. Subtracting it gives $\tau\dot{c}' + (1 + k\tau)c' = c'_0$. Define $\bar{c} = \mathcal{L}[c']$ and $\bar{c}_0 = \mathcal{L}[c'_0]$, where primes mean deviations from the reacting steady state. With zero initial deviation,

$$\frac{\bar{c}}{\bar{c}_0} = \frac{1}{\tau s + 1 + k\tau} = \frac{K_p}{\tau_r s + 1}, \qquad p = -\frac{1 + k\tau}{\tau}. \tag{11}$$

Reason: with zero initial deviation, the transform gives $[\tau s + 1 + k\tau]\bar{c} = \bar{c}_0$. Dividing isolates the output/input ratio. The pole is the value making the denominator zero. Its free-response term is e^{pt}, so a negative pole gives decay, and a more negative pole gives faster decay. Setting $s = 0$ gives $G(0) = K_p$, the settled gain for this stable lag.

Motivation and explanation

An excess of A can disappear by washout and by reaction. Both sinks strengthen the restoring effect, shortening the concentration response. At steady state, however, feed A must supply both sinks: $qc_0 = (q + kV)c$. Only the fraction $q/(q + kV)$ leaves unreacted, which explains the smaller concentration gain.

The dimensionless group $k\tau$, the Damköhler number for this model, compares the reaction timescale with the flow-through timescale. When $k\tau \ll 1$, washout dominates and $K_p \to 1$, $\tau_r \to \tau$. When $k\tau \gg 1$, little A survives, $K_p \approx 1/(k\tau)$, and $\tau_r \to 1/k$. The fluid's mean residence time is still V/q; reaction shortens the concentration memory without speeding up the liquid flow.

These conclusions require constant q, V, k and negligible thermal coupling. If flow becomes another changing input, qc contains a product of changing variables; a local model drops the product of their deviations. A prescribed new constant flow still permits a lag model, but with new coefficients. Temperature-dependent kinetics couples concentration to the energy balance. Second-order consumption kc^2 has local slope $2kc_s$, so its linearised response depends on the chosen operating concentration.

Worked example

Let $V = 2.00$ m^3, $q = 0.0200$ m^3 s^{-1}, so $\tau = 100$ s. A feed step of 3.00 mol m^{-3} gives inert-tank deviation $c'(100) = 1.89636$ mol m^{-3} and final deviation 3.00 mol m^{-3}.

With $k = 0.0100$ s^{-1}, $K_p = 0.500$, $\tau_r = 50.0$ s and $p = -0.0200$ s^{-1}. The same feed step gives final deviation 1.50 mol m^{-3} and $c'(50) = 0.948181$ mol m^{-3}.

Both responses reach 63.2121% of their own final change at their own time constant. The reacting tank reaches that fraction sooner, but its final increase is only half as large. For a common 95% settling criterion, the inert and reacting times are 299.573 s and 149.787 s respectively. Neither response finishes exactly at a finite time. The initial slopes are equal here: $\Delta c_0/\tau = K_p \Delta c_0/\tau_r = 0.0300$ mol m^{-3} s^{-1}. Reaction changes the subsequent curvature and endpoint, while the unchanged feed step sets the same initial accumulation rate.

Link to other topics

The inert-tank impulse response also describes a CSTR residence-time distribution. The same first-order structure occurs when heat transfer relaxes a sensor toward the bath temperature.

Conclusion

Flow and reaction determine the coefficient of concentration removal. That coefficient controls both the pole and the steady response to feed composition.

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PDC-05: Variable-volume tanks and finite pulse balances

Overview

Assumed: perfect mixing and the product rule.

When tank volume changes, balance the stored amount Vc, not concentration alone. A finite-volume pulse is handled by a jump balance before resuming continuous dynamics.

Intuition

Removing well-mixed liquid lowers the amount of solute but does not instantly change its concentration. Adding feed changes both solute amount and liquid volume. Forgetting the volume derivative makes these two actions look incorrectly equivalent.

Derivation

Step 1: track volume and solute. For inlet flow q_0, outlet flow q, inlet concentration c_0, tank concentration c and first-order consumption constant k, assume constant density and additive volumes:

$$\dot{V} = q_0 - q, \qquad \frac{d(Vc)}{dt} = q_0 c_0 - qc - kVc. \tag{12}$$

Reason: storage is an extensive amount. Expanding gives $V\dot{c} + c\dot{V} = q_0 c_0 - qc - kVc$; substitute $\dot{V} = q_0 - q$ and cancel $-qc$:

$$V\dot{c} = q_0(c_0 - c) - kVc. \tag{13}$$

Reason: withdrawal leaves the concentration of remaining well-mixed liquid unchanged directly. It still affects later concentration dynamics through $V(t)$.

Step 2: determine the volume history. For specified flows,

$$V(t) = V(0) + \int_0^t [q_0(\theta) - q(\theta)]\, d\theta. \tag{14}$$

Reason: integrate the volume balance first, then use the resulting V in the concentration ODE. For general time-dependent V, this is not a constant-coefficient transfer-function model. If $q_0 = 0$ and $k = 0$, the equation gives $\dot{c} = 0$ even during withdrawal.

Step 3: cross a short, finite-volume addition. A pulse adds volume V_p at concentration c_p. Let (V_-, c_-) and (V_+, c_+) denote values immediately before and after it. Neglect reaction and ordinary flow during the short addition:

$$V_+ = V_- + V_p, \qquad V_+ c_+ = V_- c_- + V_p c_p, \qquad c_+ = \frac{V_- c_- + V_p c_p}{V_- + V_p}. \tag{15}$$

Reason: total volume and solute are conserved across the jump. Use (V_+, c_+) as initial conditions for the subsequent ODE.

Why does outlet flow disappear from the concentration equation while inlet flow remains? The outlet has exactly the tank composition, so it removes solute and total volume in matching proportions. Feed at $c_0 \neq c$ changes those proportions. As a check, feed at $c_0 = c$ cannot change an inert tank's concentration even if it changes the volume. Withdrawal can still shorten the time over which a later feed change acts by reducing the stored volume.

Motivation and explanation

A Dirac impulse idealises an input with finite area and negligible duration. A finite-volume addition also changes the model's holdup; it is not automatically the same experiment as an impulse in feed concentration at fixed volume.

The model stops being valid if the tank drains, overflows, mixes poorly or has appreciable volume change on mixing. A pulse too long to neglect reaction or outlet flow must be integrated with those effects.

Worked example

A tank initially contains $V_- = 1.00$ m^3 at $c_- = 2.00$ mol m^{-3}. Add $V_p = 0.100$ m^3 at $c_p = 10.0$ mol m^{-3}. The jump balances give $V_+ = 1.100$ m^3 and $c_+ = 2.72727$ mol m^{-3}.

Treating the addition as extra solute at unchanged volume would overpredict concentration. If the addition instead displaces liquid to keep volume constant, the overflow must appear in the pulse balance.

The new concentration lies between the two original concentrations because it is their volume-weighted average. The added liquid is richer, so concentration rises, but its own volume dilutes the added solute. The jump changes the initial state for the next stage; it is not a new steady state unless later inflow, outflow and reaction also balance there.

Link to other topics

Tank-level control also begins with the volume balance. Linearising flow and concentration products later allows a local transfer-function model around a constant-volume operating point.

Conclusion

Use amount balances through volume changes and jump balances through short additions. The resulting state then determines the next continuous response.

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PDC-06: Thermocouple and heated-liquid dynamics

Overview

Assumed: lumped heat capacity and Newton's heat-transfer law.

A temperature sensor and a flowing heated tank both behave as first-order lags, but their time constants have different physical origins. Sensor delay can conceal a process change from feedback.

Intuition

A sensor must gain heat before it reports a hotter bath. Its thermal mass resists a quick change; heat transfer supplies the restoring effect. A flowing tank stores heat too, but replacement by incoming liquid provides its restoring effect in the ideal insulated model.

Derivation

Step 1: balance the sensor. Let sensor mass be m, specific heat c_p, bath temperature x, sensor temperature y, heat-transfer coefficient h and surface area A_h. Assume a uniform sensor temperature, negligible lead conduction, constant properties and a bath unaffected by the sensor:

$$mc_p\dot{y} = hA_h(x - y), \qquad \tau_m\dot{y} + y = x, \quad \tau_m = \frac{mc_p}{hA_h}. \tag{16}$$

Reason: heat inflow is proportional to bath–sensor temperature difference. Thermal capacity divided by thermal conductance has units of time. Subtract the initial equilibrium and transform:

$$G_m(s) = \frac{\bar{y}}{\bar{x}} = \frac{1}{\tau_m s + 1}, \qquad p_m = -1/\tau_m. \tag{17}$$

Reason: zero initial deviation removes the transform's initial-condition term. The pole represents relaxation of sensor temperature to bath temperature.

Step 2: balance heated liquid. For fixed tank volume V, equal flow q, density ρ, liquid heat capacity c_{pL}, feed temperature T_0 and heating rate Q, assume perfect mixing, constant liquid properties, insulation, no reaction and negligible mixer work:

$$\rho V c_{pL}\dot{T} = \rho q c_{pL}(T_0 - T) + Q, \qquad \tau\dot{T} + T = T_0 + \frac{Q}{\rho q c_{pL}}, \quad \tau = V/q. \tag{18}$$

Reason: liquid thermal storage opposes changes driven by feed and heater. For negligible pressure/volume-work effects, liquid internal-energy changes can be approximated by sensible enthalpy changes.

Subtract the steady balance and transform. Holding the other input unchanged gives

$$G_{T_0}(s) = \frac{\bar{T}}{\bar{T}_0} = \frac{1}{\tau s + 1}, \qquad G_Q(s) = \frac{\bar{T}}{\bar{Q}} = \frac{1}{\rho q c_{pL}(\tau s + 1)}. \tag{19}$$

Reason: both inputs act on the same stored energy and therefore share the same pole. Their gains differ because temperature and heating rate have different units.

In both models, normalising means dividing by the coefficient of the stored temperature, not by its derivative coefficient. For the sensor this is hA_h: at equilibrium $y = x$, so gain is one, and capacity mc_p divided by conductance hA_h gives the time constant. Increasing sensor mass slows it; increasing exposed area or heat transfer speeds it. The analogy with a mixing tank concerns this balance structure; heat transfer does not imply that sensor material is being replaced by flowing liquid.

For the heated tank, setting the derivative to zero gives $T = T_0 + Q/(\rho q c_{pL})$. A one-kelvin feed change therefore produces a one-kelvin settled tank change, whereas a watt of heating produces $1/(\rho q c_{pL})$ kelvin. Both inputs share the same relaxation rate because there is only one stored tank temperature. They do not create two separate tank states.

Motivation and explanation

The sensor model needs sufficiently rapid internal heat conduction, often assessed through a small Biot number. A thermowell, fouling or appreciable lead heat loss may add states or change the gain.

The heated-tank time constant equals residence time only under the stated assumptions. Heat loss to surroundings introduces another restoring term; changes in flow change both time constant and heater gain. A negative heater command is not physically available if the actuator only supplies heat.

Worked example

For a sensor with $m = 0.00200$ kg, $c_p = 500$ J kg^{-1} K^{-1}, $h = 200$ W m^{-2} K^{-1} and $A_h = 5.00 \times 10^{-4}$ m^2, $\tau_m = 10.0$ s. Following a bath step from 20 to 60 °C, the sensor reads 45.2848 °C after 10.0 s.

For a 1.00 m^3 water tank at $q = 0.0100$ m^3 s^{-1}, $\tau = 100$ s. With $\rho = 1000$ kg m^{-3} and $c_{pL} = 4200$ J kg^{-1} K^{-1}, a 0.420 MW heating increase gives a final 10.0 K rise and a 6.32121 K rise after 100 s. A sensor lag would delay the measured response further.

The sensor has completed the same fraction of its 40 K change that the tank has completed of its 10 K change after one respective time constant. In a measurement chain, tank temperature changes first

and supplies a changing input to the sensor. Applying the sensor's step formula directly to that gradual input would misrepresent the combined response; its dynamics must be driven by the actual tank temperature.

Link to other topics

Combining tank and sensor multiplies their transfer functions. This matters for feedback because the controller responds to the measured temperature rather than the true temperature.

Conclusion

Thermal storage divided by a restoring conductance determines each lag. The source of that conductance decides whether its time constant is a heat-transfer time or a flow residence time.

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PDC-07: Adiabatic discharge of a gas vessel

Overview

Assumed: ideal-gas properties and the open-system first law.

An insulated gas vessel cools as it discharges. The energy carried away is stream enthalpy, while the energy stored is internal energy; their difference is essential to predicting temperature and pressure.

Intuition

Insulation prevents heat exchange through the wall. It does not prevent energy loss through the outlet. The departing gas carries its internal energy plus flow work, so the remaining gas cools.

Derivation

Step 1: balance mass and energy. Let vessel volume V be fixed, stored gas mass m, outlet mass flow $w > 0$, uniform temperature T and pressure p. Assume well-mixed ideal gas with constant specific heats c_v, c_p, no inflow, wall heat storage, heat transfer or shaft work, and negligible outlet kinetic/potential energy. With $U = mc_vT$ and outlet specific enthalpy $\hat{H} = c_pT$,

$$\dot{m} = -w, \qquad \frac{d(mc_vT)}{dt} = -wc_pT. \tag{20}$$

Reason: a fixed control volume stores internal energy but stream enthalpy includes flow work at its boundary.

Step 2: eliminate time. Divide the energy balance by c_v and use $\gamma = c_p/c_v$ and $\dot{m} = -w$:

$$\frac{d(mT)}{dm} = \gamma T \quad \Rightarrow \quad T + m\frac{dT}{dm} = \gamma T \quad \Rightarrow \quad \frac{dT}{T} = (\gamma - 1)\frac{dm}{m}. \tag{21}$$

Reason: the same mass loss drives both equations, so their ratio removes the unknown discharge-flow law. Integrate from initial (m_0, T_0):

$$\frac{T}{T_0} = \left(\frac{m}{m_0}\right)^{\gamma-1}. \tag{22}$$

Step 3: obtain pressure. With mass-specific gas constant $R = c_p - c_v$, $pV = mRT$. Divide by its initial value:

$$\frac{p}{p_0} = \left(\frac{m}{m_0}\right)^{\gamma}, \qquad \frac{T}{T_0} = \left(\frac{p}{p_0}\right)^{(\gamma-1)/\gamma}. \tag{23}$$

Reason: fixed volume makes pressure proportional to mT. The result relates states along discharge, not yet to time.

Step 4: check energy reference invariance. Let specific internal energy instead be $\hat{U} = c_v T + C$, where C is a constant offset. Consistency requires $\hat{H} = \hat{U} + p/\rho = c_p T + C$. Then

$$c_v \frac{d(mT)}{dt} + C\dot{m} = -w(c_p T + C).$$

Reason: substitute $\dot{m} = -w$ and the two $-wC$ terms cancel. A changed reference cannot change the temperature prediction. Choosing unrelated enthalpy and internal-energy offsets breaks $\hat{H} - \hat{U} = RT$.

The product rule exposes the cooling mechanism directly. Expanding the stored energy and using $\dot{m} = -w$ gives $mc_v\dot{T} = -w(c_p - c_v)T = -wRT$. Removing mass at unchanged temperature would remove only its internal energy; the outlet enthalpy removes more. That excess is supplied by cooling the remaining gas. Since $w > 0$ and $T > 0$, the predicted temperature derivative is negative.

The outlet law cancels when finding temperature versus remaining mass, so different discharge rates follow the same state relation under these assumptions. They traverse it at different speeds. Although the pressure–temperature exponent resembles a reversible adiabatic closed-system relation, this derivation describes an open vessel losing mass and does not establish that the discharge is reversible.

Motivation and explanation

All temperatures in the ideal-gas and power-law relations are absolute. The relations do not provide $m(t)$ without an outlet-flow model. Constant prescribed w gives $m = m_0 - wt$ only while gas remains and the prescription is physically sustainable.

The lumped vessel model can fail during rapid discharge with spatial gradients or appreciable wall heat transfer. A nozzle's downstream static temperature need not equal vessel temperature; the outlet enthalpy/kinetic-energy accounting must match the chosen boundary.

Worked example

For $T_0 = 300$ K and $\gamma = 1.40$, discharging until $m/m_0 = 0.500$ gives $T = 227.357$ K and $p/p_0 = 0.378929$. Pressure falls faster than mass because the remaining gas also cools.

Replacing outlet enthalpy by $c_v T$ would predict constant vessel temperature under this model, which omits the outlet flow-work contribution.

Link to other topics

The boiler model also distinguishes stored internal energy from stream enthalpy. Thermodynamic reference conventions must remain consistent whenever changing mass multiplies an energy datum.

Conclusion

Insulated open-system discharge is not isothermal. A consistent mass–energy balance gives cooling and pressure decline before any particular discharge-valve law is chosen.

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PDC-08: Deviation variables and transfer-function assumptions

Overview

Assumed: basic Laplace-transform rules.

A transfer function describes the forced response of a linear, time-invariant system with zero initial state deviations. Subtracting a steady operating point allows a physically nonzero concentration or temperature to have zero deviation.

Intuition

The baseline is not the disturbance. A tank at a steady concentration already contains material; control asks how much its concentration changes when an input changes. Deviation variables remove the baseline while retaining exactly the perturbation we care about.

Derivation

Step 1: subtract the steady state. For an inert mixed tank, $\tau\dot{c} + c = c_0$. At equilibrium $c_s = c_{0s}$. Define $c' = c - c_s$ and $c_0' = c_0 - c_{0s}$. Substitution gives

$$\tau\dot{c}' + c_s + c' = c_{0s} + c_0' \quad \Rightarrow \quad \tau\dot{c}' + c' = c_0'. \tag{24}$$

Reason: the steady values satisfy the original balance and cancel. A prime denotes deviation; a dot denotes time differentiation. They are different operations.

Step 2: transform with the initial condition retained. For any deviation output y', write

$$\bar{y}(s) = \mathcal{L}[y'] = \int_0^\infty y'(t)e^{-st}\,dt, \qquad \mathcal{L}[\dot{y}'] = s\bar{y} - y'(0^-). \tag{25}$$

Reason: integration by parts produces the initial-value term. The Laplace variable s has inverse-time units. Transformed variables generally have their original units multiplied by time.

For $\tau\dot{y}' + y' = K_p x'$,

$$\bar{y} = \frac{K_p}{\tau s + 1}\bar{x} + \frac{\tau y'(0^-)}{\tau s + 1}. \tag{26}$$

Reason: collect $\bar{y}$ and isolate it. The first term is the input-driven response; the second is the free response to stored initial deviation. Set all initial state deviations to zero to define

$$G(s) = \left.\frac{\bar{y}}{\bar{x}}\right|_{\text{zero initial state deviations}} = \frac{K_p}{\tau s + 1}. \tag{27}$$

Step 3: state the scope. Constant coefficients and superposition define the linear, time-invariant model used here. A SISO transfer function relates one input–output pair. With multiple inputs, hold the other deviations at zero when defining each transfer-matrix element.

Reason: a raw ratio of output to one input with other inputs or initial-state effects present does not isolate that pair's dynamics.

Subtracting a baseline and neglecting an initial condition are separate decisions. Baseline subtraction is an exact change of coordinates. Zero initial deviation is justified only by actually starting at that baseline. If a tank begins above its chosen equilibrium, that stored excess relaxes even when the input deviation stays zero; the second term in equation (26) accounts for it. A transfer function isolates how the imposed input acts, so that free response is handled separately.

For a stable lag, the denominator $\tau s + 1$ is the transformed competition between storage and removal. The transform has made a differential equation algebraic; it has not added a new physical mechanism. Returning to the time domain gives the same exponential relaxation derived by an integrating factor.

Motivation and explanation

Subtraction gives zero initial deviations only when the process actually starts at the chosen equilibrium. If it starts elsewhere, include the free-response term. For a second-order ODE, both output and derivative initial conditions matter.

Zero initial state does not require a step input to have zero value or derivative immediately after $t = 0$. A step deliberately jumps. Taking $t = 0$ at the disturbance is a convenient origin, not a special physical assumption. A nonlinear model only gains a local transfer function after linearisation.

Worked example

An inert tank initially has $c_s = c_{0s} = 5.00$ kmol m^{-3}. Its feed steps to 8.00 kmol m^{-3}. The transformed input deviation is $\bar{c}_0 = 3.00/s$, not $8.00/s$, and the initial output deviation is zero.

For $\tau = 100$ s,

$$\bar{c} = \frac{3.00}{s(100s + 1)}, \qquad c(t) = 5.00 + 3.00(1 - e^{-t/100}).$$

The absolute concentration after 100 s is 6.89636 kmol m^{-3}. The same transfer function would apply around any inert-tank baseline at unchanged flow and volume.

Link to other topics

Deviation variables support both first-order response analysis and Taylor linearisation. A pole belongs to the process transfer function; additional factors in an input transform belong to the imposed signal.

Conclusion

A transfer function separates process dynamics from baseline, initial conditions and input waveform. Preserve that distinction when interpreting its poles and gain.

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PDC-09: First-order step and impulse responses

Overview

Assumed: the transfer function of a self-regulating mixed or thermal tank.

For $G(s) = K_p/(\tau s + 1)$ with $\tau > 0$, gain sets the final response and time constant sets the speed. Step and impulse experiments expose different features of the same stored-state dynamics.

Intuition

A persistent step keeps supplying a new condition until the tank settles. A brief pulse supplies a finite perturbation and then leaves the tank to relax. The response to the pulse therefore decays, while the step response accumulates that decaying response.

Derivation

Step 1: apply a step. Let $H(t)$ be the unit Heaviside step and $x' = \Delta x H(t)$. For zero initial state, $\bar{x} = \Delta x/s$, so

$$\bar{y} = \frac{K_p \Delta x}{s(\tau s + 1)} = K_p \Delta x \left[\frac{1}{s} - \frac{1}{s + 1/\tau}\right]. \tag{28}$$

Reason: multiply the proposed partial fractions by $s(\tau s + 1)$ to check that their numerator is one. Invert each term:

$$y'(t) = K_p \Delta x (1 - e^{-t/\tau}), \quad \dot{y}'(t) = \frac{K_p \Delta x}{\tau} e^{-t/\tau}. \tag{29}$$

Reason: the pole $-1/\tau$ generates a decaying exponential; the input's $1/s$ generates the constant final term.

Step 2: interpret the time constant. At $t = \tau$, the normalised response is $1 - e^{-1}$. The initial tangent is $y' \approx (K_p \Delta x/\tau)t$, which meets the final response at $t = \tau$. To reach fraction f of the final change, solve $f = 1 - e^{-t/\tau}$:

$$t_f = -\tau \ln(1 - f), \qquad 0 < f < 1. \tag{30}$$

Reason: the remaining distance is an exponentially decaying fraction.

Step 3: apply an impulse. Let $\delta(t)$ be a unit-area Dirac impulse, with $\mathcal{L}[\delta] = 1$. It is a distribution, not an ordinary finite-height signal. For impulse area I_x, $x' = I_x\delta(t)$ and

$$\bar{y} = I_x G(s), \qquad y'(t) = \frac{K_p I_x}{\tau} e^{-t/\tau} \quad (t \geq 0). \tag{31}$$

Reason: invert the single first-order pole. Since δ has inverse-time units, I_x has input units times time. Integrating the ODE across the impulse gives $\tau[y'(0^+) - y'(0^-)] = K_p I_x$, confirming the output jump.

The distance still to travel is $K_p\Delta x - y' = K_p\Delta x\, e^{-t/\tau}$. In each time interval of length τ, this remaining distance is multiplied by e^{-1}; the response never uses up a fixed amount per second. That is why the rise is initially steep and then flattens. The initial tangent reaches the final level at τ only if its starting slope were maintained; the actual slope immediately starts falling.

An impulse has a different endpoint because its forcing ends. Its finite area changes the stored state across the brief input, then the unforced state decays back to baseline. A step continually supports a displaced equilibrium. Their different final values reflect different experiments, not different process gains.

Motivation and explanation

A lag begins responding immediately; its delayed approach is not dead time. Negative K_p reverses the response direction but does not change stability when $\tau > 0$. Comparing the magnitude of gain with one makes sense only after accounting for units or using a consistent scaling.

The unit-step derivative is the unit-impulse response for this zero-state LTI system. Its impulse-response area is K_p. Real pulses approximate the impulse only when their duration is short relative to the dynamics and they do not appreciably change the process parameters.

Worked example

With $K_p = 2.00$, $\tau = 20.0$ s and step $\Delta x = 3.00$ in input units, the final output change is 6.00 and $y'(20) = 3.79272$ in output units. For a unit-area input pulse, $y'(20) = 0.0367879$ in the corresponding output units.

A first-order response reaches 63.2121% of its final change at one time constant, 95.0213% at three and 99.3262% at five. For $\tau = 100$ s, the exact time to 95% is 299.573 s.

For comparison, at two time constants the step has completed 86.4665% of its final change. All these fractions refer to the output change from its initial value, not to its absolute value. Dividing by $K_p\Delta x$ removes amplitude and units, leaving one common curve against t/τ for every stable first-order lag without delay.

Link to other topics

An integrator responds to a step with continuing accumulation instead of settling. Cascading several lags changes the early response and creates higher-order dynamics.

Conclusion

The step measures steady sensitivity and relaxation speed; the impulse isolates the decaying response to a finite input area.

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PDC-10: Integrators and transport dead time

Overview

Assumed: volume balances and the time-shift rule for Laplace transforms.

Two elementary process elements behave very differently from a self-regulating lag. An integrator stores a persistent imbalance; ideal transport dead time shifts a signal without smoothing it.

Intuition

A tank with a fixed pumped outlet cannot increase outflow automatically when its level rises. Extra inflow keeps accumulating. A plug-flow pipe instead carries a temperature front downstream: nothing arrives until its transit time has elapsed, then the same front appears.

Derivation

Step 1: derive the integrator from level. For a tank of constant cross-sectional area A, level h, inlet q_0 and prescribed pumped outlet q, assume both flows are independent of h:

$$A\dot{h} = q_0 - q, \qquad A\dot{h}' = q_0' - q', \qquad \bar{h} = \frac{1}{As}(\bar{q}_0 - \bar{q}). \tag{32}$$

Reason: volume is Ah; subtract the balanced baseline and transform with zero initial level deviation. With net-flow deviation $x' = q_0' - q'$, the transfer function is $1/(As)$, with pole at zero. A constant imbalance Δq gives $h' = \Delta q\, t/A$; an impulse of net volume I_V gives a persistent level jump I_V/A .

A scaled form is $T_H \dot{y}' = x'$ and $G = 1/(T_H s)$. Here T_H is a holdup time when x', y' are defined with compatible normalised units; A in the dimensional level equation is an area, not a time.

Step 2: derive transport delay. For pipe length L, uniform constant velocity u, negligible axial mixing and heat exchange, a fluid element takes $t_d = L/u$ to reach the outlet. With inlet temperature deviation x' and outlet y', zero deviation history before the disturbance gives

$$y'(t) = \begin{cases} 0, & t < t_d, \\ x'(t - t_d), & t \geq t_d. \end{cases} \tag{33}$$

Reason: each outlet parcel carries its inlet condition from one transit time earlier. Substitute $\theta = t - t_d$ into the transform:

$$\bar{y} = \int_{t_d}^{\infty} x'(t - t_d) e^{-st}\, dt = e^{-st_d} \int_{0}^{\infty} x'(\theta) e^{-s\theta}\, d\theta, \qquad G_d(s) = e^{-st_d}. \tag{34}$$

Reason: the exponential factor accounts for the time shift and does not change the signal's eventual amplitude.

The missing term in the level balance is a level-dependent removal term. A lag would have a contribution proportional to $-h'$ that grows as level rises and eventually cancels the changed inflow. Here the prescribed pump removes the same flow at every level, so no new level restores the balance. When the extra inflow ends, the level stops rising but retains its accumulated offset. This is the physical memory represented by $1/s$.

A delay stores a history of conditions travelling along a pipe. It does not continuously blend that history into a single mixed state. For a step, compare the shapes: a lag begins immediately and curves toward a final value; an integrator ramps while the imbalance persists; an ideal delay stays unchanged and then jumps. Increasing t_d shifts the arrival time without reducing step amplitude or rounding the front.

Motivation and explanation

The integrator is not bounded-input bounded-output (BIBO) stable: a bounded nonzero constant input creates an unbounded ramp. Its zero-input level deviation remains constant, so it is neutrally stable in that state sense rather than an exponentially growing mode. It is not asymptotically stable or self-regulating. Real level is bounded by emptying or overflowing, where the ideal model stops applying.

A pure delay is BIBO stable and has unit steady gain. It is not a finite-order rational transfer function. A first-order lag begins changing immediately; a pure delay waits and then reproduces the input. Axial dispersion, wall heat exchange or velocity variation makes a real pipe more than a pure delay.

Worked example

For $A = 2.00$ m^2 and persistent net inflow 0.00200 m^3 s^{-1}, level rises at 0.00100 m s^{-1} and increases by 0.0600 m in 60.0 s.

For a 20.0 m pipe of diameter 0.0400 m carrying 0.00100 m^3 s^{-1}, cross-sectional velocity is 0.795775 m s^{-1} and transit time is 25.1327 s. An ideal inlet temperature step reaches the outlet after that time with its original amplitude.

Link to other topics

A gravity-draining tank introduces a level-dependent outflow and can become self-regulating. Transport delay matters in controller design because measurements may describe a condition created well before the current actuator command.

Conclusion

A pole at zero represents accumulation without restoring action. The exponential delay factor represents travel time, not a slow approach to equilibrium.

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PDC-11: Second-order models, poles and damping regimes

Overview

Assumed: two coupled first-order balances and exponential solutions of linear ODEs.

Second-order behaviour requires two dynamic states or one second-order equation. Its pole pair determines whether deviations relax monotonically, decay while oscillating, persist, or grow.

Intuition

Two tanks smooth a disturbance twice. A mass and spring exchange kinetic and elastic energy and can overshoot. Both have two states, but the mechanism of coupling decides whether oscillation is possible.

Derivation

Step 1: eliminate an intermediate tank state. Let x' be inlet deviation, y' the first-tank outlet and z' the second, with positive time constants τ_1, τ_2 and unity gains:

$$\tau_1 \dot{y}' + y' = x', \qquad \tau_2 \dot{z}' + z' = y'.$$

Reason: each mixed-tank balance relaxes its state toward the upstream condition. Substitute $y' = \tau_2 \dot{z}' + z'$ and its derivative into the first balance:

$$\tau_1 \tau_2 \ddot{z}' + (\tau_1 + \tau_2)\dot{z}' + z' = x'. \tag{35}$$

Step 2: define the standard form. For a general ODE $a_2 \ddot{y}' + a_1 \dot{y}' + a_0 y' = b x'$ with constant coefficients and $a_0 \neq 0$, the conventional form applies when $a_2/a_0 > 0$:

$$\tau^2 \ddot{y}' + 2\zeta\tau \dot{y}' + y' = K_p x', \quad \tau^2 = a_2/a_0, \quad 2\zeta\tau = a_1/a_0, \quad K_p = b/a_0. \tag{36}$$

Reason: divide by a_0 and define $\tau > 0$ as the second-order time scale and ζ as the dimensionless damping factor. For a forced mass–spring system $m\ddot{y}' + \mu\dot{y}' + k y' = F'$, $\tau = \sqrt{m/k}$, $\zeta = \mu/(2\sqrt{mk})$ and force-to-displacement gain is $1/k$. Weight has already cancelled when displacement is measured from equilibrium.

Step 3: transform and find the poles. Zero initial state and derivative give

$$G(s) = \frac{K_p}{\tau^2 s^2 + 2\zeta\tau s + 1}, \qquad p_{1,2} = \frac{-\zeta \pm \sqrt{\zeta^2 - 1}}{\tau}. \tag{37}$$

Reason: solve the quadratic characteristic equation; each root multiplies time in a free-response exponential.

Damping	Pole pattern	Response of this constant-numerator model
$\zeta > 1$	Two negative real poles	Overdamped; monotonic stable step
$\zeta = 1$	Repeated pole $-1/\tau$	Critically damped
$0 < \zeta < 1$	Complex pair with negative real part	Decaying oscillations
$\zeta = 0$	Pair $\pm i/\tau$	Undamped; persistent free oscillations
$\zeta < 0$	Poles with positive real part	Growing deviations

Reason: the real part controls decay/growth and the imaginary part supplies oscillation frequency. At $\zeta = -1$, the repeated positive pole also gives instability.

Normalising by a_0 makes the final output/input ratio visible as $K_p = b/a_0$; matching the derivative coefficients then defines τ and ζ. Do not read a_1 alone as a first-order time constant: both derivative terms jointly determine two poles. For a complex pair, $e^{(-\zeta/\tau + i\omega_d)t}$ separates into decay envelope $e^{-\zeta t/\tau}$ and oscillation. The envelope time is τ/ζ, while the oscillation period depends on ω_d.

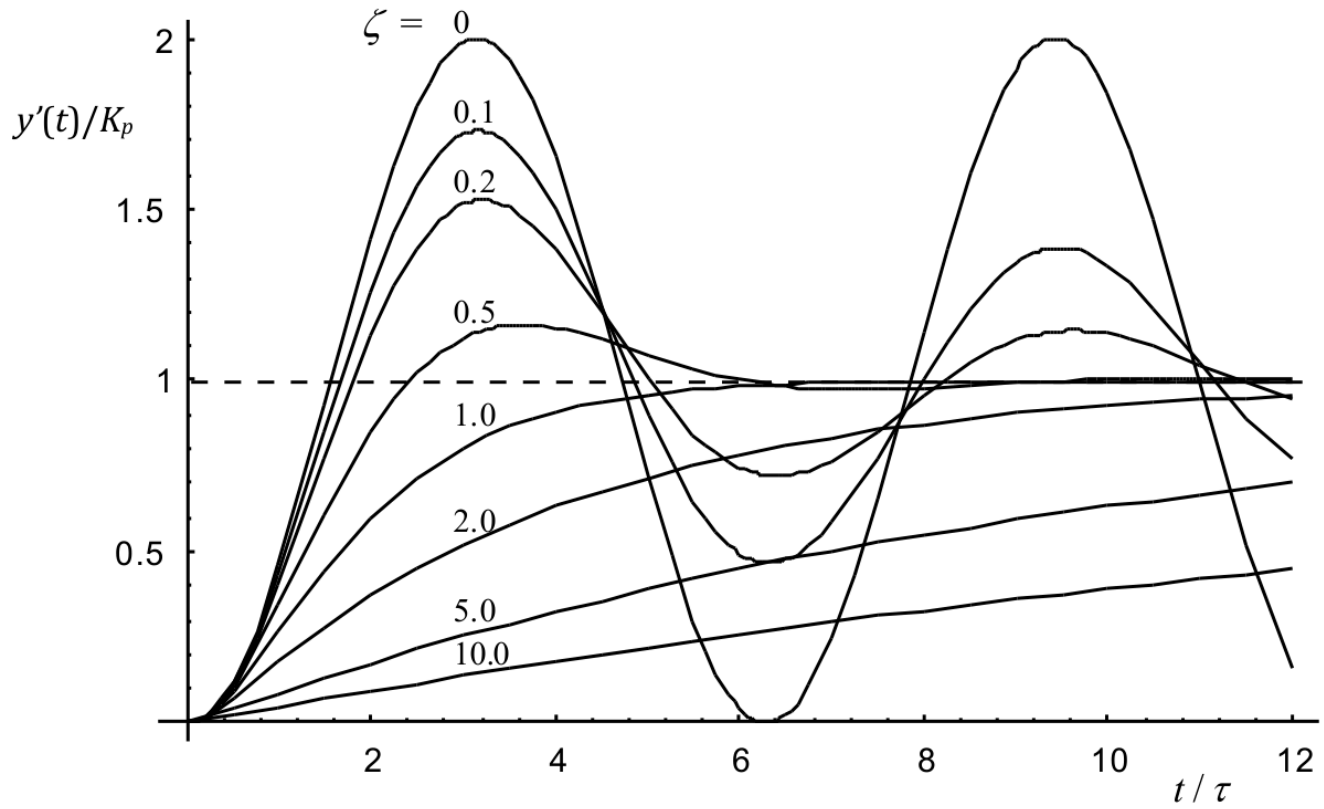
For the mass–spring analogy, displacement can keep increasing after crossing equilibrium because velocity is still positive: kinetic energy carries it past the target. Dissipation reduces that stored energy on each swing. Two sequential mixing tanks lack this inertial exchange mechanism; the downstream tank merely follows a progressively changing upstream input. The shared second-order equation does not make their physical couplings identical.

Motivation and explanation

The natural angular frequency is $\omega_n = 1/\tau$, in rad per unit time. The undamped period is $2\pi\tau$; τ itself is not an oscillation period.

For two passive lags, $\tau = \sqrt{\tau_1\tau_2}$ and $\zeta = (\tau_1 + \tau_2)/(2\sqrt{\tau_1\tau_2}) \geq 1$, because $(\sqrt{\tau_1} - \sqrt{\tau_2})^2 \geq 0$. Thus a simple cascade of two positive lags cannot be underdamped. Equal lags are critically damped. Undamped poles are not BIBO stable, even though their free oscillations are bounded.

Worked example



The graph uses t/τ horizontally and output normalised by its unit-step gain vertically. For two tanks with $\tau_1 = 20.0$ s and $\tau_2 = 80.0$ s, the combined time scale is 40.0 s and $\zeta = 1.25$. The poles are -0.0500 and -0.0125 s^{-1}: both modes decay without oscillation.

For $\tau = 10.0$ s and $\zeta = 0.500$, poles are $-0.0500 \pm 0.0866025i$ s^{-1}. The response oscillates with a decaying envelope rather than behaving like the two passive tanks.

Read the response graph in two parts: the common final normalised value expresses gain, while the curvature and crossings express damping. For the unequal tanks, the pole nearer zero, -0.0125 s^{-1}, corresponds to the slower 80.0 s mode and governs the late tail when its coefficient is nonzero. For the complex example, the real part sets how quickly peaks shrink and the imaginary part sets how closely they are spaced.

Link to other topics

Pole analysis generalises to higher-order models. Overshoot and decay ratio translate this second-order pole pair into directly observable response features.

Conclusion

Second order describes the number of dynamic states; damping describes how their modes behave. The physical coupling determines which damping regimes are possible.

PDC-12: Second-order step response, overshoot and decay ratio

Overview

Assumed: the standard second-order model with constant numerator, no zeros or dead time, and zero initial output and derivative.

Deriving the step response makes overshoot, oscillation period and decay ratio consequences of the pole pair. These relations apply to the standard model, not every process with two poles.

Intuition

The steady response is a constant particular solution. The transient is the free mode needed to match the starting state. In an underdamped process, that free mode repeatedly crosses the final value while losing amplitude.

Derivation

Step 1: separate steady and transient parts. Let the input step be Δx and normalise $r(t) = y'(t)/(K_p \Delta x)$, with nonzero $K_p \Delta x$. Then $\tau^2 \ddot{r} + 2\zeta\tau\dot{r} + r = 1$. For distinct poles, write $r = 1 + C_1 e^{p_1 t} + C_2 e^{p_2 t}$. Initial conditions require

$$C_1 + C_2 = -1, \qquad p_1 C_1 + p_2 C_2 = 0.$$

Reason: neither output nor its first derivative can jump under a finite step for this strictly proper second-order model. Solving these two equations gives

$$r = 1 + \frac{p_2 e^{p_1 t} - p_1 e^{p_2 t}}{p_1 - p_2}. \tag{38}$$

Step 2: handle the repeated root. At $\zeta = 1$, $p = -1/\tau$ and the transient has form $(C_1 + C_2 t)e^{-t/\tau}$. Enforcing $r(0) = \dot{r}(0) = 0$ gives $C_1 = -1$, $C_2 = -1/\tau$, so

$$r = 1 - (1 + t/\tau)e^{-t/\tau}. \tag{39}$$

Reason: repeated roots need a polynomial factor; simply setting the two exponential coefficients equal would lose one independent solution.

Step 3: use complex poles. For $0 < \zeta < 1$, define $a = \zeta/\tau$ and $\omega_d = \sqrt{1 - \zeta^2}/\tau$. Write $r = 1 + e^{-at}(C \cos \omega_d t + D \sin \omega_d t)$. The same initial conditions give $C = -1$ and $D = -a/\omega_d$,

hence

$$r = 1 - e^{-\zeta t/\tau}\left[\cos(\omega_d t) + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin(\omega_d t)\right]. \tag{40}$$

Reason: conjugate modes combine into a real damped sinusoid.

Step 4: locate the first peak. Differentiate the preceding result; cosine terms cancel:

$$\dot{r} = \frac{e^{-\zeta t/\tau}}{\tau\sqrt{1-\zeta^2}}\sin(\omega_d t).$$

Reason: extrema occur at zero sine. The first peak is at $t_p = \pi/\omega_d$, where cosine is -1. If $B = |K_p \Delta x|$ is the total steady change and A is the first peak's excess magnitude above the final value,

$$\frac{A}{B} = \exp\left[-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}\right], \qquad t_p = \frac{\pi\tau}{\sqrt{1-\zeta^2}}. \tag{41}$$

Step 5: compare successive peaks on the same side. They are separated by $T = 2\pi/\omega_d$. If C is the next same-side peak's excess magnitude, its envelope has fallen by e^{-aT}:

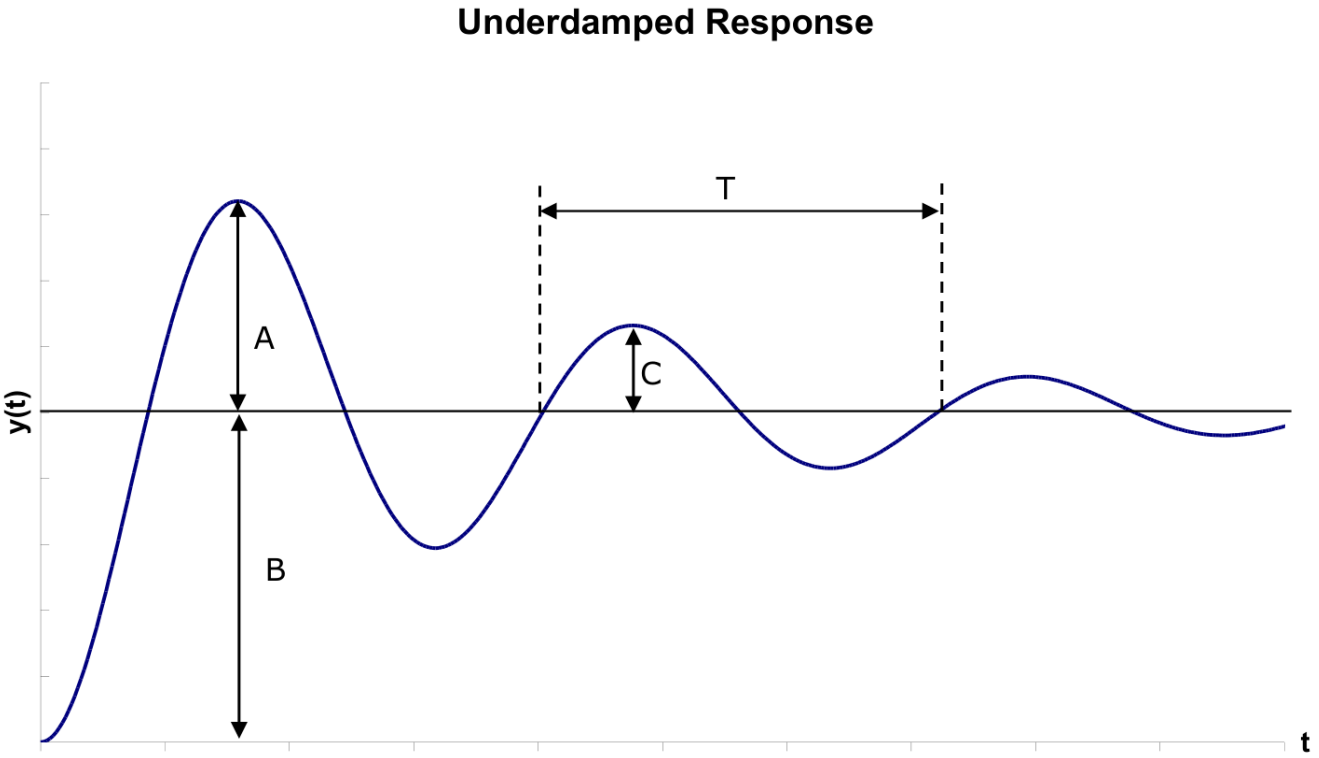
$$\frac{C}{A} = \exp\left[-\frac{2\pi\zeta}{\sqrt{1-\zeta^2}}\right] = \left(\frac{A}{B}\right)^2, \qquad T = \frac{2\pi\tau}{\sqrt{1-\zeta^2}}. \tag{42}$$

Reason: compare peaks measured relative to the final value, not their absolute heights. Consecutive extrema on opposite sides are only half a period apart.

The constant 1 in the normalised solution is the forced equilibrium. The transient starts at -1 to cancel it at $t = 0$ and must also match the zero initial slope. This is why the oscillatory part is not an arbitrary sine curve placed around the endpoint: its sine and cosine weights are fixed by the starting state. As t grows, the transient decays and leaves the forced equilibrium.

At the first peak the response has travelled half an oscillation period. There the sine term is zero and the cosine is -1, so the excess above the final value equals the exponential decay factor at that instant. Successive peaks on the same side are a full period apart, so their excesses shrink by the full-period decay factor. In the figure, measure excesses from the final-value line; measuring peak heights from zero would mix the steady part into the transient ratio.

Motivation and explanation



The horizontal reference through the oscillations represents the final value; B spans the initial-to-final change. For negative gain, reflect the response and use magnitudes in the overshoot ratio.

Taking $\zeta \to 0$ gives $r = 1 - \cos(t/\tau)$, with no final settled value. Taking $\zeta \to 1$ requires the repeated-root limit rather than substituting into a singular-looking expression. Zeros or delays alter measured response shape and invalidate a blind fit of these formulas.

Worked example

For $\tau = 10.0$ s and $\zeta = 0.500$, $\omega_d = 0.0866025$ rad s^{-1}, $t_p = 36.2760$ s and $T = 72.5520$ s. Overshoot is $A/B = 0.163034$, or 16.3034%, and same-side decay ratio is $C/A = 0.0265799$. A unit-gain unit-step response therefore first peaks at 1.16303.

For critical damping, the normalised response at $t = \tau$ is $1 - 2e^{-1} = 0.264241$. Unlike a first-order lag, it has zero initial slope.

The 16.3034% overshoot means the first peak exceeds the final change by that fraction; it does not mean the output peaks at 0.163034. The same-side ratio 0.0265799 means the next excess is only that fraction of the first excess. The repeated-lag response's zero initial slope also explains its smaller progress at $t = \tau$: the second tank must wait for the first tank to develop a temperature difference, even though there is no exact waiting interval.

Link to other topics

Observed oscillation can reveal a complex pole pair. Cascaded tank examples illustrate the critically damped formula without requiring a mechanical analogy.

Conclusion

Overshoot comes from the transient mode carrying the output past its steady value. Pole decay and frequency determine the excess peak and the spacing between peaks.

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PDC-13: Series blocks and the pipe–two-tank response

Overview

Assumed: a plug-flow delay and perfectly mixed tank energy balances.

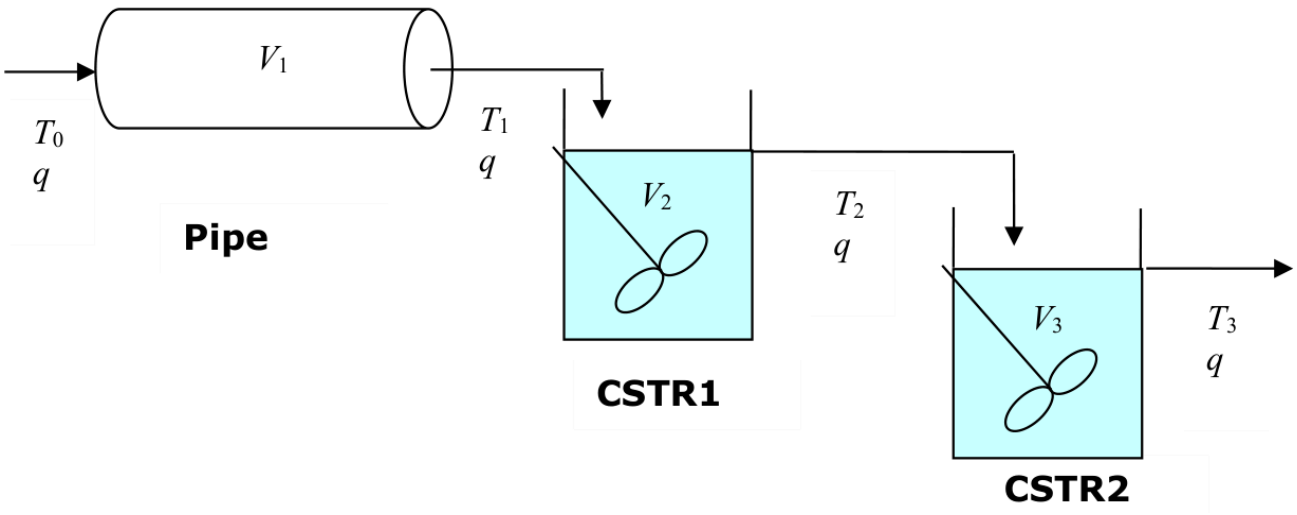
A pipe followed by two insulated tanks combines transport delay with two first-order lags. Multiplying blocks preserves each element's physical role and produces a delayed second-order response.

Intuition

The pipe postpones the arrival of a hot front. The first tank dilutes it into stored liquid; the second smooths that already changing temperature. The outlet stays unchanged until the front arrives, then initially changes very gently.

Derivation

Step 1: construct each element. For pipe length L, diameter D, constant volumetric flow q and uniform velocity $u = 4q/(\pi D^2)$, the delay is $t_d = L/u$. Let temperatures along the sequence be T_0, T_1, T_2, T_3, and tank volumes be V_2, V_3.



For ideal plug flow, $\bar{T}_1 = e^{-t_d s}\bar{T}_0$. With constant liquid properties and no heat losses, each tank balance is $\rho V_i c_p \dot{T}_i = \rho q c_p (T_{i-1} - T_i)$, so its deviation transfer function is $1/(\tau_i s + 1)$, $\tau_i = V_i/q$.

Reason: the pipe preserves the input waveform after transit, whereas tank storage gives a lag.

Step 2: substitute through the series. For blocks G_1, G_2, G_3, $\bar{T}_1 = G_1\bar{T}_0$, $\bar{T}_2 = G_2\bar{T}_1$, $\bar{T}_3 = G_3\bar{T}_2$. Therefore

$$\frac{\bar{T}_3}{\bar{T}_0} = \frac{e^{-t_d s}}{(\tau_2 s + 1)(\tau_3 s + 1)}. \tag{43}$$

Reason: the output of one element is the next element's input; repeated substitution multiplies their SISO transfer functions.

Step 3: invert equal lags. For $\tau_2 = \tau_3 = \tau$ and inlet step ΔT_0, first remove the delay and decompose

$$\frac{1}{s(\tau s + 1)^2} = \frac{1}{s} - \frac{\tau}{\tau s + 1} - \frac{\tau}{(\tau s + 1)^2}.$$

Reason: multiplying by the common denominator verifies the coefficients. The inverse transforms of the last two terms are $e^{-t/\tau}$ and $(t/\tau)e^{-t/\tau}$. Restore the delay by shifting the complete causal response:

$$T_3'(t) = \begin{cases} 0, & t < t_d, \\ \Delta T_0 \left[1 - \left(1 + \frac{t - t_d}{\tau}\right) e^{-(t - t_d)/\tau}\right], & t \geq t_d. \end{cases} \tag{44}$$

Reason: the pipe postpones the entire two-tank response. The rational part has a repeated pole $-1/\tau$ and is critically damped.

The delay and lags answer different questions. The delay says when the response can first appear. The two tank states say how it develops after that arrival. Immediately after arrival, tank 1 starts warming, but tank 2 still has almost no inlet–contents temperature difference, so its initial slope is zero. For every positive elapsed time after arrival the outlet does change; the initially flat tangent does not add another interval of dead time.

Multiplying the zero-frequency gains gives one: with no heat loss, every element ultimately reaches the new inlet temperature. The repeated pole supplies the $te^{-t/\tau}$ term, which changes the rise shape without changing that endpoint. Shift both occurrences of time in the causal response when reinstating the pipe delay; shifting only the exponential would violate the starting conditions.

Motivation and explanation

Scalar LTI series multiplication commutes, so reordering these ideal blocks leaves the overall input–output transfer function unchanged. Their intermediate signals do change. Reordering real equipment may also change heat losses, mixing, flows or the relevant model; the algebra does not establish physical interchangeability.

Two lags produce a zero initial slope after the delay. They create a smooth rise, not extra exact dead time. The final gain remains one because the insulated system eventually passes the changed

temperature through unchanged.

Worked example

Let $q = 1.00 \times 10^{-3}$ m^3 s^{-1}, $D = 0.0400$ m, $L = 20.0$ m and $V_2 = V_3 = 0.100$ m^3. The calculated velocity is 0.795775 m s^{-1}, $t_d = 25.1327$ s and both time constants are 100 s. Thus

$$G(s) = \frac{e^{-25.1327 s}}{(100 s + 1)^2}.$$

For a 1.00 K inlet step, $T_3'(125) = 0.263753$ K. Rounding the transit time to 25 s is reasonable for a coarse estimate, but the calculation here retains the value from the specified geometry.

Link to other topics

Series-block algebra extends to sensor and actuator lags. Parallel paths instead require signed addition of contributions to an output.

Conclusion

Build the overall response from physically derived elements. The delay sets its onset; the repeated tank pole sets its subsequent smooth approach.

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PDC-14: Summing contributions and duplicating signals

Overview

Assumed: superposition in linear deviation models.

Series elements multiply; independent contributions to the same output add with their signs. A signal branch copies information and does not divide its value like a material-flow splitter.

Intuition

In a heated tank, warmer feed and extra heater power both affect temperature. Each input passes through its own response path, then the temperature contributions combine. Measuring that temperature does not consume part of it.

Derivation

Step 1: begin with the balance. For fixed V, q, ρ, c_p, no heat loss or reaction, set $\tau = V/q$ and $C_f = \rho q c_p$, the flowing heat-capacity rate. The tank deviation equation is

$$\tau \dot{T}' + T' = T_0' + Q'/C_f. \tag{45}$$

Reason: subtract the steady energy balance so the two forcing contributions remain. Transform with zero initial deviation:

$$\bar{T} = G_1 \bar{T}_0 + G_2 \bar{Q}, \qquad G_1 = \frac{1}{\tau s + 1}, \quad G_2 = \frac{1}{C_f(\tau s + 1)}. \tag{46}$$

Reason: both sources feed the same state, but the heater input must first be converted from power to temperature-equivalent forcing.

Step 2: apply a signed summing junction. A junction labelled $+$ on one path and $-$ on another forms $\bar{y} = \bar{y}_1 - \bar{y}_2$. If both paths start with the same input $\bar{x}$, their equivalent block is $G_1 - G_2$; two positive paths give $G_1 + G_2$.

Reason: superposition is addition of output effects, including their signs. Signals summed at a junction must have compatible units; raw temperature and power cannot be added.

Step 3: duplicate a signal. If one ideal branch sends $\bar{T}$ onward and another to a sensor, both pre-sensor branches equal $\bar{T}$. The measured signal is $\bar{T}_m = G_m \bar{T}$, where G_m is the sensor transfer function.

Reason: a branch copies a numerical variable. It is not a material balance and does not halve temperature. Sensor dynamics or loading, if relevant, must be represented separately.

It is legitimate to sum the two temperature contributions because each is already expressed in output-temperature units. The factor $1/C_f$ converts heating power into the feed-temperature change with the same thermal effect. Then the common lag applies to their sum. A block diagram is bookkeeping for these equations: a line is a signal, a summing point adds signed effects, and a branch sends the same value to more than one destination.

Exact dynamic compensation requires the paths to cancel as functions of s, not merely at $s = 0$. If the heater has an extra actuator lag, its final temperature contribution can still cancel the cold feed while its early contribution is too slow. That produces a temporary temperature drop even when the eventual temperature error is zero.

Motivation and explanation

Superposition holds for the linear model at fixed coefficients. If flow changes, the heater gain and time constant change too; products such as $q'T'$ require linearisation before this simple summation is justified.

For an ideal measurement scale, $G_m(0) = 1$, but a sensor may still lag. Treating the measured output as instantly equal to the true output hides a pole in the feedback path.

Worked example

For water with $C_f = 42000$ W K^{-1} and $\tau = 100$ s, let the feed temperature fall by 5.00 K while heating rises by 210000 W. Their step contributions sum to

$$T'(t) = \left[-5.00 + \frac{210000}{42000}\right](1 - e^{-t/100}) = 0.$$

The exact cancellation follows because both paths share the same tank lag in this model and the commands occur together. An actuator lag in the heat path, a delayed disturbance measurement, or inaccurate C_f produces a transient error even when the steady effects cancel.

Link to other topics

Feedforward exploits signed summation to oppose disturbance effects. Transfer-function matrices organise the same idea when several inputs affect several outputs.

Conclusion

Write the signal equations before simplifying a block diagram. Their units, signs and branch meanings prevent most combination errors.

PDC-15: Transfer-function matrices and coupled CSTR concentrations

Overview

Assumed: constant-volume, isothermal CSTR balances for $A \to B$.

A MIMO model is a matrix of input–output responses. The reacting CSTR shows how an input can affect its own species directly and another species through a reaction pathway.

Intuition

More A in the feed raises reactor A and therefore makes more B. More B in the feed raises B without creating A because the reaction is irreversible. The matrix should express this directional coupling, not artificially give every input the same effect.

Derivation

Step 1: define the matrix convention. With n input deviations and m output deviations,

$$\bar{\boldsymbol{y}} = \boldsymbol{G}(s)\bar{\boldsymbol{x}}, \qquad \bar{y}_i = \sum_{j=1}^{n} G_{ij}(s)\bar{x}_j. \tag{47}$$

Reason: output row i sums the contributions from input columns j. Define each G_{ij} with all other input deviations and initial state deviations zero. Each entry has units of its row's output divided by its column's input.

Step 2: balance both components. Let V be volume, q equal inlet/outlet flow, $\tau = V/q$, $k > 0$ first-order rate constant, c_{A0}, c_{B0} feed concentrations and c_A, c_B tank concentrations. Assume constant density, one mole B formed per mole A consumed, no volume change and perfect mixing:

$$\tau \dot{c}_A + (1 + k\tau)c_A = c_{A0}, \qquad \tau \dot{c}_B + c_B = c_{B0} + k\tau c_A. \tag{48}$$

Reason: A reaction loss is an equal B reaction source. Subtract the steady balances and transform:

$$\bar{c}_A = \frac{\bar{c}_{A0}}{\tau s + 1 + k\tau}, \qquad \bar{c}_B = \frac{\bar{c}_{B0}}{\tau s + 1} + \frac{k\tau \bar{c}_{A0}}{(\tau s + 1)(\tau s + 1 + k\tau)}.$$

Reason: solve the A equation first, then substitute it into the B equation. This produces

$$\begin{bmatrix} \bar{c}_A \\ \bar{c}_B \end{bmatrix} = \begin{bmatrix} \dfrac{1}{\tau s + 1 + k\tau} & 0 \\ \dfrac{k\tau}{(\tau s + 1)(\tau s + 1 + k\tau)} & \dfrac{1}{\tau s + 1} \end{bmatrix} \begin{bmatrix} \bar{c}_{A0} \\ \bar{c}_{B0} \end{bmatrix}. \tag{49}$$

Step 3: check the conserved total. Add the two ODEs:

$$\tau \frac{d(c_A + c_B)}{dt} + (c_A + c_B) = c_{A0} + c_{B0}. \tag{50}$$

Reason: reaction cancels in the total concentration balance. Equivalently, the A-feed column sums to $1/(\tau s + 1)$, because

$$\frac{1}{\tau s + 1 + k\tau} + \frac{k\tau}{(\tau s + 1)(\tau s + 1 + k\tau)} = \frac{1}{\tau s + 1}.$$

Read the first column as one experiment: change feed A, hold feed B unchanged, and observe both outputs. The upper entry is direct survival of A; the lower entry is production of B through the changing A inventory. Read the second column as a different experiment: feed B changes without changing A. The zero in the upper-right entry records the absence of a reverse reaction or other coupling under these assumptions.

Adding the two output rows checks more than the final numbers. Reaction redistributes molecules between A and B but cannot alter their total in this one-to-one conversion. The total therefore follows the inert-tank mode at all times. A faster A-removal mode is compatible with a slower total-concentration mode because consuming A creates B rather than removing total A plus B.

Motivation and explanation

The two dynamic modes have poles $-(1 + k\tau)/\tau$ and $-1/\tau$. The B response to feed A contains both modes: A first changes, then its changed conversion supplies B. The A response to feed B is zero under the stated irreversible, uncoupled chemistry.

This is a MIMO concentration model. The heated tank with two inputs but only one temperature output is MISO, a single-output special case. If temperature affects k, or multiple reactions occur, the zero entry and pole structure can change.

Worked example

Take $\tau = 100$ s and $k = 0.0100$ s^{-1}. A unit step in c_{A0} with no c_{B0} change gives

$$c'_A = \tfrac{1}{2}(1 - e^{-2t/100}), \qquad c'_B = \tfrac{1}{2} - e^{-t/100} + \tfrac{1}{2}e^{-2t/100}.$$

These follow by taking the A response and subtracting it from total response $1 - e^{-t/100}$. At 100 s, $c'_A = 0.432332$ and $c'_B = 0.199788$ in the concentration unit used for the input. Final deviations

are 0.500 each. Their sum tracks an inert-tank concentration response at every time.

The final half-and-half split is about the extra feed A: half survives as A and half appears as B. Early on, A begins rising immediately, while B initially has zero slope because its extra production rate first needs extra A to build up. Thus the equal final gains do not imply equal transient curves.

Link to other topics

Material conservation is a strong check on a coupled model. The boiler matrix will combine two different balances and include an integrating inventory output.

Conclusion

Rows identify outputs and columns identify inputs. Coupling entries should follow actual pathways, and conserved combinations provide independent verification.

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PDC-16: Boiler mass–energy balance and integrating inventory

Overview

Assumed: saturated-phase properties, a rigid control volume and simultaneous linear equations.

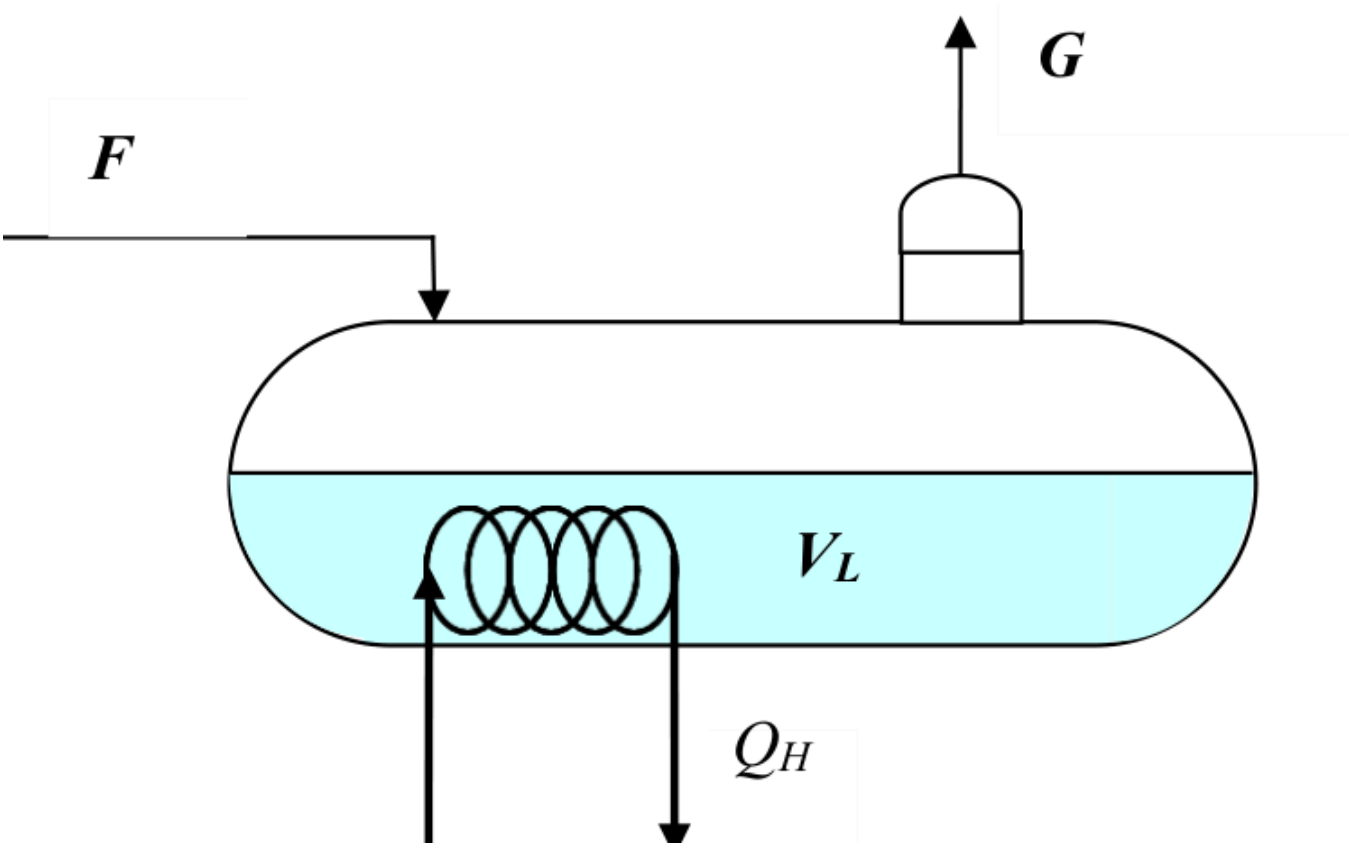
For a boiler held at fixed pressure, mass and energy balances determine liquid-inventory dynamics and steam production together. This idealised model contains an integrator for inventory and algebraic feed/heat effects on steam flow.

Intuition

Extra heat evaporates stored liquid unless the feed rises to replace it. Extra subcooled feed can temporarily reduce steam output because some heat must warm that feed. Pressure is assumed already maintained; its dynamic control is outside this reduced model.

Derivation

Step 1: define properties and storage. Let F be liquid-feed mass flow, G steam mass flow, Q_H heat input, V_L, V_G liquid/vapour volumes, and $V_T = V_L + V_G$ fixed vessel volume. Hold pressure at 8 bar absolute, so saturation temperature is 170.4 °C and phase properties remain constant. Densities are ρ_L, ρ_G; specific internal energies are $\hat{U}_L, \hat{U}_G$; feed and steam specific enthalpies are $\hat{H}_F, \hat{H}_G$. Ignore heat loss and other work.



Mass is $M = \rho_L V_L + \rho_G (V_T - V_L)$. Internal energy is $U = \rho_L \hat{U}_L V_L + \rho_G \hat{U}_G (V_T - V_L)$. Differentiate at fixed properties:

$$(\rho_L - \rho_G)\dot{V}_L = F - G, \qquad (\rho_L \hat{U}_L - \rho_G \hat{U}_G)\dot{V}_L = F\hat{H}_F - G\hat{H}_G + Q_H. \tag{51}$$

Reason: total vessel volume is constant, so increasing liquid volume displaces vapour volume. Enthalpy accounts for energy carried by streams; internal energy describes storage.

Step 2: eliminate steam flow. Define $D_\rho = \rho_L - \rho_G$ and $D_U = \rho_L \hat{U}_L - \rho_G \hat{U}_G$. From mass, $G = F - D_\rho \dot{V}_L$. Substitute into energy:

$$D_U \dot{V}_L = F(\hat{H}_F - \hat{H}_G) + D_\rho \hat{H}_G \dot{V}_L + Q_H.$$

Reason: use one balance to close the unknown flow in the other. Define

$$B = D_\rho \hat{H}_G - D_U = \rho_L(\hat{H}_G - \hat{U}_L) - \rho_G(\hat{H}_G - \hat{U}_G).$$

Then

$$\dot{V}_L = aF + bQ_H, \quad G = cF + dQ_H, \quad a = \frac{\hat{H}_G - \hat{H}_F}{B},\ b = -\frac{1}{B},\ c = 1 - D_\rho a,\ d = \frac{D_\rho}{B} \tag{52}$$

Reason: isolate $\dot{V}_L$, then recover G from mass. With zero initial deviation, subtract the steady balances and transform:

$$\begin{bmatrix} \bar{V}_L \\ \bar{G} \end{bmatrix} = \begin{bmatrix} a/s & b/s \\ c & d \end{bmatrix} \begin{bmatrix} \bar{F} \\ \bar{Q}_H \end{bmatrix}. \tag{53}$$

Reason: liquid volume is the integral of its rate; steam flow is algebraic in this reduced model.

The signs follow the energy pathways. At fixed feed, extra heat evaporates liquid, so the heat-to-liquid-volume coefficient b is negative and the heat-to-steam coefficient d is positive. At fixed heat, extra subcooled feed requires warming, so it accumulates as liquid and can reduce steam production. A feed-to-steam coefficient slightly below zero is therefore compatible with mass conservation: the changed inventory absorbs more mass than the feed increase alone.

There is no V_L term on the right of $\dot{V}_L = aF + bQ_H$. Consequently, a persistent nonzero $aF' + bQ'_H$ cannot be removed merely by reaching a different level. The integrator represents that continuing accumulation. Steam flow has no separate stored state in this fixed-pressure approximation; the algebraic balance assigns it an immediate change. Real pressure and heat-transfer dynamics would resolve that idealisation into additional dynamics.

Motivation and explanation

The inventory pole is at zero: at fixed F, Q_H the model provides no level-dependent restoring effect. A persistent mismatch drains or fills the boiler until the assumed phase inventory no longer exists. The algebraic steam-flow row represents direct feedthrough, not a lag with an extremely short measured time constant.

At steady state $G_s = F_s$ and $Q_{Hs} = F_s(\hat{H}_G - \hat{H}_F)$. The model deliberately omits pressure dynamics, finite evaporation kinetics and heat storage in equipment. It is valid only while saturated liquid and vapour coexist and the fixed-pressure approximation is adequate.

Worked example

Use $\hat{H}_F = 675.5$, $\hat{H}_G = 2768.3$, $\hat{U}_L = 720.0$, $\hat{U}_G = 2576.0$ kJ kg^{-1}, and $\rho_L = 896.9$, $\rho_G = 4.2$ kg m^{-3}. Calculate $B = 1.83631261 \times 10^{6}$ kJ m^{-3}. With F, G in kg s^{-1} and Q_H in kJ s^{-1} (kW),

$$\begin{bmatrix} \bar{V}_L \\ \bar{G} \end{bmatrix} \approx \begin{bmatrix} 1.13968 \times 10^{-3}/s & -5.44570 \times 10^{-7}/s \\ -0.0173881 & 4.86137 \times 10^{-4} \end{bmatrix} \begin{bmatrix} \bar{F} \\ \bar{Q}_H \end{bmatrix}.$$

Here s is in s^{-1}. At fixed heat, a 1.00 kg s^{-1} feed increase gives $\dot{V}'_L = 1.13968 \times 10^{-3}$ m^3 s^{-1} and $G' = -0.0173881$ kg s^{-1}. Increasing heat simultaneously by 2092.8 kW cancels the inventory drift and gives $G' = 1.00$ kg s^{-1}, as both steady balances require.

If heat is expressed in watts instead, divide the heat-column coefficients by 1000. The feed enthalpy includes warming the inlet from 160 °C to saturation; using only saturated-liquid latent heat would miss that load.

Link to other topics

Inventory control restores a stable level when a process naturally integrates. The boiler also shows why proper transfer functions can contain direct input-to-output terms.

Conclusion

Mass and energy jointly determine steam flow and inventory. The integrating liquid-volume row identifies the need for inventory regulation; the algebraic row reflects the fixed-pressure idealisation.

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PDC-17: Poles, repeated modes and stability

Overview

Assumed: partial fractions and complex exponentials.

Poles describe the free dynamic modes visible in an input–output model. Repeated poles add polynomial factors; complex conjugate poles combine into real oscillations. Stability must be defined carefully at poles on the imaginary axis.

Intuition

After a pulse is over, each dynamic mode retains a memory of the perturbation. A negative real part makes that memory fade. A positive real part amplifies it. Zero real part does not supply decay, even when a particular free response stays bounded.

Derivation

Step 1: identify process poles. For a rational SISO model $G(s) = N(s)/D(s)$ in reduced form, poles are roots of $D(s)$ and zeros are roots of $N(s)$. In $\bar{y} = G(s)\bar{x}$, extra roots supplied by $\bar{x}$ are input-signal poles, not additional process states.

Step 2: invert elementary pole terms. For a pole p and integer $j \geq 1$,

$$\mathcal{L}^{-1}\left[\frac{C_j}{(s-p)^j}\right] = C_j \frac{t^{j-1}}{(j-1)!} e^{pt}. \tag{54}$$

Reason: $\mathcal{L}[e^{pt}] = 1/(s-p)$ and successive powers of t arise by differentiating with respect to s. Repeated poles therefore require all powers from $1/(s-p)$ through $1/(s-p)^j$ in the partial fraction expansion.

Step 3: combine a complex pair. Let $p = a + ib$, residue $C = \alpha + i\beta$, and conjugate terms p^*, C^*, where $i^2 = -1$. Euler's formula gives

$$Ce^{pt} + C^* e^{p^* t} = 2e^{at}[\alpha \cos(bt) - \beta \sin(bt)]. \tag{55}$$

Reason: imaginary terms cancel because a real physical model has conjugate poles and residues. The envelope is controlled by a, and frequency by $|b|$.

Step 4: relate modes to stability. For a finite-dimensional proper rational model with no unstable hidden modes, strictly negative real parts imply asymptotic decay and BIBO stability. A right-half-plane pole gives instability. Simple imaginary-axis poles give nondecaying free modes but fail BIBO

stability; bounded resonant forcing can accumulate indefinitely. Repeated imaginary-axis poles can grow even without forcing.

Reason: multiplying a decaying exponential by any finite-degree polynomial still tends to zero; without exponential decay that guarantee disappears.

For a simple real pole, moving farther left increases the decay rate because e^{pt} falls more quickly. A complex pair adds repeated sign changes while its real part still controls the envelope. Repeating a negative pole adds a factor such as t, so a mode can initially grow in magnitude before decaying; the exponential ultimately wins. A repeated pole at zero instead gives powers of time with no decaying exponential to suppress them.

Among observable stable modes with nonzero weights, poles nearest the imaginary axis usually govern the late tail. Their weights still matter: a slow mode with a very small coefficient may be barely visible over the time interval of interest. Pole locations predict possible rates and frequencies, while the particular input, output and numerator determine how much each mode appears.

Motivation and explanation

Cancellation can hide a mode from one input–output transfer function. An exact unstable pole–zero cancellation is not evidence of internal stability: an initial condition or another input/output path may expose the hidden growing state. Use the full state model when making that claim.

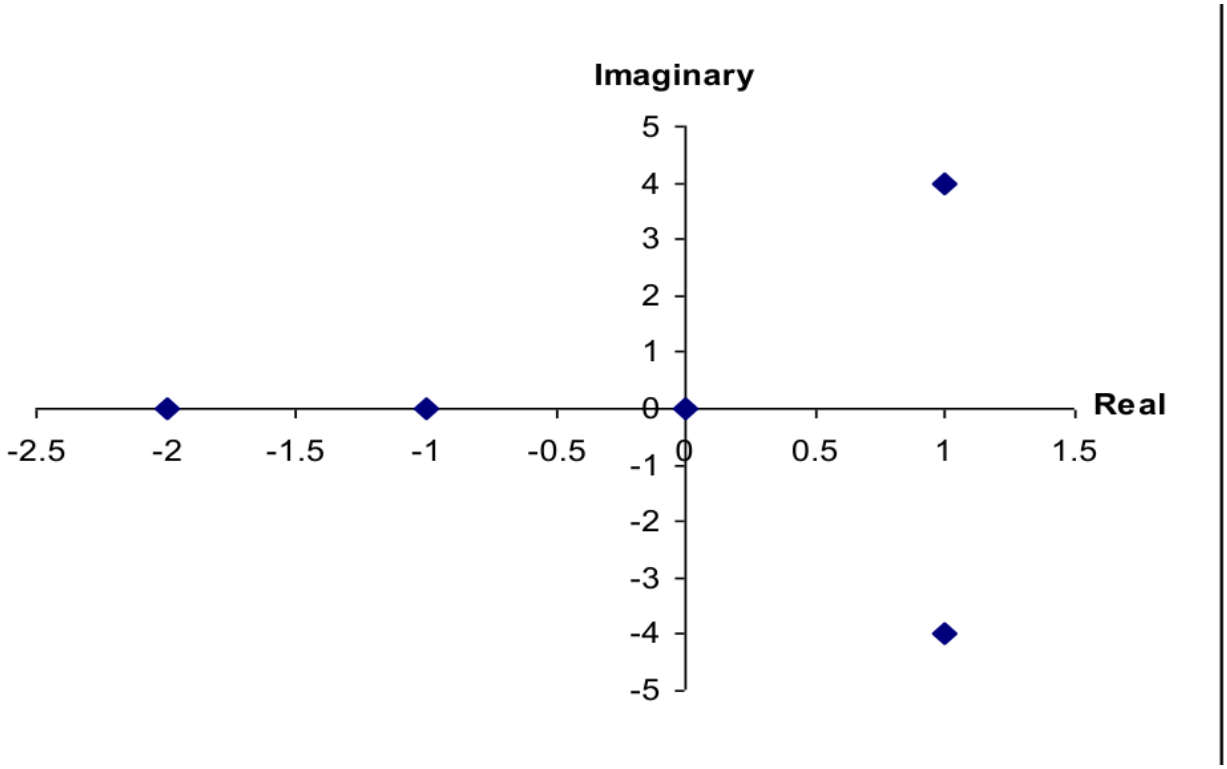
For physical rational models, properness requires numerator degree no larger than denominator degree. Equal degrees allow direct feedthrough. Strictly proper models have lower numerator degree. Pure transport delay is causal but non-rational, so the polynomial-degree test does not describe it.

Worked example

Consider the dimensional time coordinate in seconds and the transfer function

$$G(s) = \frac{1}{s^6 + 2s^5 + 14s^4 + 60s^3 + 81s^2 + 34s} = \frac{1}{s(s+1)^2(s+2)[(s-1)^2 + 16]}.$$

The poles are 0, -1 twice, -2 and $1 \pm 4i$, in s^{-1}. The positive-real-part pair makes the system unstable.



For a unit-area impulse, partial fractions give the response for $t \geq 0$:

$$y'(t) = \frac{1}{34} - \frac{e^{-2t}}{50} - \frac{e^{-t}}{100} - \frac{te^{-t}}{20} + \frac{e^{t}[\cos(4t) - 0.25\sin(4t)]}{1700}.$$

The constant comes from the process pole at zero because the input is an impulse. The te^{-t} term reveals the repeated pole, and the growing oscillation reveals $1 \pm 4i$. The residues and factorisation can be checked by recombining the rational terms before inversion.

The growing oscillation may start with a small coefficient, but its e^{t} envelope eventually overwhelms the decaying terms and constant. A short observation that looks quiet cannot overturn the positive-real-part pole test. Here the constant term comes from a process integrator; in a stable lag's step response, a constant can instead come from the step's input pole. Keeping the experiment specified avoids confusing those cases.

Link to other topics

Pole locations determine whether the final-value theorem can be used. Zeros influence the weighting of these modes and can cause an initial response opposite to the final change.

Conclusion

Interpret each pole as a mode with a decay rate and, when complex, an oscillation frequency. Distinguish visible input–output stability from stability of all internal states.

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PDC-18: Zeros, direct feedthrough and inverse response

Overview

Assumed: poles and partial-fraction responses.

Zeros reshape the forced response without creating a new exponential mode. A right-half-plane zero can make the output initially move opposite to its final direction, even when every process pole is stable.

Intuition

Two physical pathways can compete. A fast pathway may initially push the output down while a slower but stronger pathway eventually raises it. Their sum can therefore dip before rising; the zero encodes that competition.

Derivation

Step 1: separate modes from their weights. Write a strictly proper model with distinct poles as $G(s) = \sum_j C_j/(s - p_j)$. For each pole,

$$C_j = \lim_{s \to p_j} (s - p_j) G(s). \tag{56}$$

Reason: multiplying removes that term's simple singularity; all other terms vanish in the limit. Changing the numerator changes residues C_j, hence the relative amplitudes of modes, while unreduced denominator roots stay fixed unless cancellation occurs.

Step 2: build an inverse-response example. Add two stable first-order pathways:

$$G(s) = \frac{4}{s+1} - \frac{6}{s+2} = \frac{2(1-s)}{(s+1)(s+2)}. \tag{57}$$

Reason: put the contributions over a common denominator. Poles are $-1, -2$ and the zero is $+1$. Both modes decay, but the zero is in the right half-plane. For a unit step,

$$\bar{y} = \frac{G(s)}{s} = \frac{1}{s} - \frac{4}{s+1} + \frac{3}{s+2}, \qquad y' = 1 - 4e^{-t} + 3e^{-2t}. \tag{58}$$

Reason: invert each partial fraction. At zero, $y' = 0$ but $\dot{y}' = 4 - 6 = -2$; the final value is 1. The initial direction opposes the final direction.

Step 3: distinguish feedthrough from storage. If numerator and denominator degrees are equal, long division gives $G(s) = D_0 + G_{sp}(s)$, where D_0 is a constant and G_{sp} is strictly proper. In the time domain, $y' = D_0 x'$ plus the stored-state response.

Reason: a constant block passes an input directly and may create an immediate output jump under a step. It need not be unphysical: the reduced boiler steam-flow model has this form.

The two pathways settle to different strengths: their gains are 4 and -3, so the positive contribution wins at long times. Initially, their slopes are 4 and -6, so the faster negative path wins. That change of dominance explains the dip without requiring an unstable pole. The zero at $+1$ appears when the two pathway expressions are combined; it describes their cancellation in the transfer numerator, not an extra growing state.

Feedthrough is another distinct feature. With a finite step and strictly proper lag pathways, the state output remains continuous and only its slope changes. A nonzero direct term D_0 adds an immediate jump $D_0 \Delta x$ on top of the stored-state response. Such a jump, a gradual inverse response, and a transport waiting time therefore have different mathematical and physical causes.

Motivation and explanation

A right-half-plane zero does not by itself make the open-loop transfer function unstable. It constrains response shaping and makes exact stable causal inversion problematic. A numerator zero can cancel a denominator pole in a mathematical expression, but that can hide internal dynamics.

Not every stable two-pole model has the monotonic step response of the standard constant-numerator second-order model. Its zero structure and feedthrough matter too. The example assumes linearity and no transport delay.

Worked example

For the two-pathway model, solve $\dot{y}' = 4e^{-t} - 6e^{-2t} = 0$. This gives the minimum at $t = \ln(3/2) = 0.405465$ s, where $y' = -1/3 = -0.333333$. The response eventually approaches $+1$.

Physically, the negative pathway has the faster pole -2 and initially contributes slope -6; the positive pathway contributes slope $+4$. Their final step contributions are -3 and $+4$, leaving a positive total.

Link to other topics

The final-value theorem predicts the endpoint but cannot reveal inverse response. Controller design must account for both poles and zeros when interpreting the transient.

Conclusion

Poles set the available modes; zeros set how the input excites them. A stable process can still have an initially misleading response direction.

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PDC-19: Steady-state gain and the final-value theorem

Overview

Assumed: Laplace transforms of step inputs.

Steady-state gain is the change in settled output per change in settled input. Evaluating $G(0)$ gives this gain only when the relevant response actually settles; an algebraic limit alone cannot prove that it does.

Intuition

A constant input eventually leaves only a constant balance if all transient modes decay. Gain is the sensitivity of that remaining balance. If a mode grows or keeps oscillating, there is no settled output to interpret.

Derivation

Step 1: read the gain from a balance. For $\tau\dot{y}' + y' = K_p x'$, a final constant input Δx and settled output Δy satisfy $\Delta y = K_p \Delta x$.

Reason: a steady state has zero derivative. Gain has output/input units; it need not be dimensionless.

Step 2: use the transform. The final-value theorem states, under its validity conditions,

$$\lim_{t\to\infty} y'(t) = \lim_{s\to 0} s\bar{y}(s). \tag{59}$$

For a proper rational signal transform, a standard sufficient condition is that all poles of $s\bar{y}$ lie strictly in the left half-plane after cancellations. Equivalently, $\bar{y}$ may have a simple pole at the origin for a constant final value, but no other nondecaying modes. A finite causal delay preserves a settling signal's endpoint.

Reason: the theorem requires decay of every nonconstant term. Check the complete output transform, including the imposed input and any initial-state contribution, rather than only one process denominator.

Step 3: specialise to a step. For zero-state $\bar{y} = G(s)\Delta x/s$,

$$\Delta y = \Delta x \lim_{s\to 0} G(s), \qquad K_p = G(0), \tag{60}$$

provided the step response settles and the limit is finite.

Reason: the step's $1/s$ factor cancels the theorem's factor s. For a stable MIMO model, applying a step in one input at a time gives the gain matrix $\boldsymbol{G}(0)$. Integrating output paths generally lack a finite steady gain.

Evaluating at $s = 0$ removes derivative factors, which is why it reproduces the static balance when a static balance is actually reached. The final-value theorem makes that connection precise; it does not supply decay that the process lacks. A unit step has transform $1/s$, so $s\bar{y} = G(s)$ for that experiment, and the stability condition must be checked before taking its limit.

Gain and speed can change independently. Multiplying a stable transfer function by a constant scales its endpoint without moving its poles. Adding pure delay shifts its timing without changing $G(0)$. Moving a pole while adjusting the numerator can change speed at fixed gain. A final-value calculation deliberately discards transient detail and must be paired with a dynamics check when timing matters.

Motivation and explanation

A lag's pole comes from the process; a simple $1/s$ in the output transform may merely come from a step input. Confusing them would label a settling step response unstable.

A pure transport delay satisfies $e^{-t_d s} \to 1$ at $s = 0$. It changes timing but not steady gain. In an unstable model, a finite $G(0)$ does not establish an attainable endpoint.

Worked example

Take

$$G(s) = \frac{10e^{-5s}}{(s+1)(s+5)(s+10)}, \qquad x'(t) = 1 - e^{-t/10}, \quad \bar{x} = \frac{1}{s(10s+1)}.$$

The nonconstant modes have negative poles $-1, -5, -10, -0.1$, and the delay is finite. Hence

$$y'(\infty) = \lim_{s \to 0} \frac{10e^{-5s}}{(10s+1)(s+1)(s+5)(s+10)} = 0.200.$$

For contrast, $G(s) = 1/(s-1)$ with a unit step gives $y' = e^t - 1$. The algebraic value $G(0) = -1$ is finite, but the positive pole prevents a final value; using the theorem to claim $y'(\infty) = -1$ would be invalid.

Link to other topics

Linearisation produces local gains that are derivatives of nonlinear steady relations. Controller decisions require that endpoint information together with transient poles, zeros and delay.

Conclusion

Check settling conditions before evaluating the endpoint. Once they hold, the final-value theorem avoids solving an entire transient just to find its final change.

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PDC-20: Local linearisation and reaction-order comparison

Overview

Assumed: Taylor expansion, partial derivatives and reactor balances.

Linearisation replaces a nonlinear model by its tangent dynamics at one operating point. The steady state cancels, while the local slopes become gain and decay coefficients. The approximation is valid for sufficiently small deviations.

Intuition

A curved relationship can look almost straight over a narrow interval. Control near a fixed set point uses that local slope, not a single straight line for every possible operating condition. Moving the operating point can change both sensitivity and response speed.

Derivation

Step 1: define linearity. An operator F is linear if $F(au + bv) = aF(u) + bF(v)$ for constants a, b. Powers and products of changing variables generally violate this property. A coefficient that depends on time may preserve linearity but loses time invariance.

Step 2: expand a dynamic model. Let output/state y obey $\dot{y} = f(y, x)$ for input x. Choose a constant equilibrium (y_s, x_s) with $f(y_s, x_s) = 0$; define $y' = y - y_s$, $x' = x - x_s$. Taylor expansion gives

$$\dot{y}' \approx f(y_s, x_s) + f_y|_s\, y' + f_x|_s\, x' = f_y|_s\, y' + f_x|_s\, x'. \tag{61}$$

Reason: the constant term vanishes at equilibrium, and terms quadratic or higher in deviations are neglected. Subscripts on f denote partial derivatives, evaluated at the operating point. With additional disturbance d, add $f_d|_s d'$; with several states, use the matrix of partial derivatives.

Step 3: expose the local transfer function. Write $a = f_y|_s$, $b = f_x|_s$. For zero initial deviation,

$$G(s) = \frac{b}{s - a}. \quad \text{If } a < 0: \quad \tau = -1/a, \quad K_p = -b/a. \tag{62}$$

Reason: transform $\dot{y}' = ay' + bx'$ and isolate the output. The local restoring slope a determines the pole. If $a = 0$, the first-order model is an integrator and higher-order terms may be decisive for nonlinear stability.

Step 4: compare reaction orders. For fixed q, V, T and $\tau = V/q$, first-order consumption gives $\tau \dot{c}_A + (1 + k_1\tau)c_A = c_{A0}$, exactly linear in these concentration variables.

Second-order consumption gives $\tau \dot{c}_A + c_A + k_2\tau c_A^2 = c_{A0}$. At its equilibrium, $c_{A0s} = c_{As} + k_2\tau c_{As}^2$. Substitute $c_A = c_{As} + c'_A$:

$$\tau \dot{c}'_A + (1 + 2k_2\tau c_{As})c'_A + k_2\tau (c'_A)^2 = c'_{A0}.$$

Reason: expand the square and cancel the steady balance. Neglect the final quadratic term to obtain

$$G(s) \approx \frac{1}{\tau s + 1 + 2k_2\tau c_{As}}, \quad \tau_{lin} = \frac{\tau}{1 + 2k_2\tau c_{As}}, \quad K_p = \frac{1}{1 + 2k_2\tau c_{As}}. \qquad (63)$$

Reason: the derivative of the reaction sink is $2k_2 c_{As}$, not $k_2 c_{As}$.

To see what the tangent approximation removes, expand $(c_{As} + c'_A)^2 = c_{As}^2 + 2c_{As}c'_A + (c'_A)^2$. The first term belongs to the equilibrium and cancels; the second changes the local removal slope; the third is the curvature being neglected. A higher positive operating concentration therefore gives a stronger local reaction sink, a shorter time constant and a smaller feed-to-concentration gain. This has the same storage-versus-removal interpretation as the first-order reacting tank, but its coefficients now depend on the operating point.

For a general product of two changing variables, $(q_s + q')(c_s + c') = q_s c_s + q_s c' + c_s q' + q'c'$. Keep both terms linear in deviations and drop only the final product. Calling the perturbations small is meaningful because their product is smaller in order than either individual deviation; it does not justify discarding one of the linear contributions.

Motivation and explanation

The approximation discards the curvature, not the physical meaning of the original balance. Second-order k_2 has units concentration^{-1} time^{-1}; k_1 has inverse-time units. Constant-temperature first-order reaction is linear, but a temperature-dependent Arrhenius rate introduces nonlinear coupling if temperature is a changing state.

The neglected quadratic reaction term is small relative to its first-order reaction perturbation when $|c'_A| \ll 2c_{As}$. Near $c_{As} = 0$, that particular comparison fails and one must inspect the full balance instead. A Taylor model says little about distant equilibria or large excursions.

Worked example

Let $\tau = 100$ s, $k_2 = 0.0100$ m^3 mol^{-1} s^{-1} and $c_{As} = 1.00$ mol m^{-3}. The corresponding feed equilibrium is $c_{A0s} = 2.00$ mol m^{-3}. Linearisation gives $\tau_{lin} = 33.3333$ s and $K_p = 0.333333$.

At unchanged operating conditions, a sufficiently small feed-composition increase has one-third of that increase in its final reactor concentration. Linearising about a different c_{As} gives different

coefficients; this is a local model rather than a globally linear reaction.

The local gain is also the slope of the steady relation. Differentiating $c_{A0s} = c_{As} + k_2 \tau c_{As}^2$ gives $dc_{As}/dc_{A0s} = 1/(1 + 2k_2 \tau c_{As})$. Thus the dynamic linearisation and the tangent to the equilibrium curve predict the same small final change. A finite change follows the curved relation, so this slope need not give its exact endpoint.

Link to other topics

The same Taylor procedure converts a gravity-outflow tank into a lag. Products of valve flow and concentration require it in the three-tank mixing process.

Conclusion

Linearise only after finding a consistent equilibrium. The local derivatives identify the restoring mode and steady sensitivity of small disturbances.

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PDC-21: Gravity-draining tank: linear model and exact response

Overview

Assumed: a volume balance and square-root outflow from Bernoulli's law.

Unlike a tank with prescribed pumped withdrawal, a gravity-draining tank has a restoring outflow. Linearisation makes it a local first-order lag; solving the nonlinear step reveals the approximation's limits.

Intuition

A higher level increases the driving head and therefore increases outflow. That feedback in the physics opposes accumulation without a controller. Because outflow increases as a square root, its slope weakens at larger levels; a single tangent model cannot predict every new equilibrium exactly.

Derivation

Step 1: write and linearise the balance. Let area be A, level $h > 0$, inlet flow q_0, and outlet $q = k\sqrt{h}$ with fixed valve/outlet coefficient $k > 0$. Assume constant density and area, quasi-steady outflow, no overflow, and no other inflows. Then

$$A\dot{h} = q_0 - k\sqrt{h}, \qquad q_{0s} = k\sqrt{h_s}. \tag{64}$$

Reason: increasing level stores volume and increases the gravity-driven loss. Define deviations and expand $\sqrt{h_s + h'} \approx \sqrt{h_s} + h'/(2\sqrt{h_s})$. Cancel the steady terms:

$$A\dot{h}' = q_0' - \frac{k}{2\sqrt{h_s}}h', \qquad \tau\dot{h}' + h' = K_p q_0', \quad \tau = \frac{2A\sqrt{h_s}}{k}, \quad K_p = \frac{2\sqrt{h_s}}{k}. \tag{65}$$

Reason: the outflow slope supplies a restoring term. The transfer function is $G = K_p/(\tau s + 1)$, with pole $-k/(2A\sqrt{h_s})$. Gain has units s m^{-2}, equivalent to level per volumetric-flow change.

Step 2: compare new equilibria. After a step Δq_0, the local model gives

$$h'_{lin} = K_p \Delta q_0 (1 - e^{-t/\tau}), \qquad h'_{exact,\infty} = K_p \Delta q_0 + (\Delta q_0/k)^2. \tag{66}$$

Reason: the exact equilibrium is $h_\infty = [(q_{0s} + \Delta q_0)/k]^2$; subtract $h_s = (q_{0s}/k)^2$ and expand the square. The missing quadratic term reveals the finite-step error.

Step 3: integrate the exact nonlinear step. Let $z = \sqrt{h}$, $z_0 = \sqrt{h_s}$ and $z_f = (q_{0s} + \Delta q_0)/k$. Since $\dot{h} = 2z\dot{z}$,

$$2Az\dot{z} = k(z_f - z), \qquad dt = \frac{2A}{k}\,\frac{z}{z_f - z}\,dz.$$

Reason: substituting the square-root state makes the equation separable. Use $z/(z_f - z) = -1 + z_f/(z_f - z)$ and integrate from z_0:

$$t = \frac{2A}{k}\left[z_0 - z + z_f \ln\left|\frac{z_f - z_0}{z_f - z}\right|\right]. \tag{67}$$

Reason: the logarithm comes from integrating $1/(z_f - z)$ with its negative derivative. This implicit exact solution applies along a positive-level trajectory before reaching the new equilibrium; a zero step simply gives constant h. For a positive inlet step, $z_0 < z < z_f$.

Step 4: linearise in the transformed state too. Write $z = z_0 + z'$ in $2Az\dot{z} = q_0 - kz$. To first order,

$$2Az_0\dot{z}' + kz' = q_0', \qquad h' = 2z_0 z' + O((z')^2).$$

Reason: $z'\dot{z}'$ is a second-order perturbation, and the output transformation must also be expanded. Together these recover exactly the earlier local h' model. Squaring the approximate $z(t)$ without dropping $(z')^2$ gives a different finite-step approximation, not the first-order linear level model.

The outlet's local slope is $dq/dh|_s = k/(2\sqrt{h_s})$. Dividing area by this slope gives the time constant; the reciprocal of the outflow slope gives the steady level gain. More area slows the response without changing the eventual level needed to pass a given flow. At greater operating depth, the outflow slope is smaller, so an extra inlet flow needs a larger level rise and that rise develops more slowly. These trends follow from the square-root law, not from an assumed universal tank time constant.

For either positive or negative physically admissible inlet changes, the exact final level differs from the tangent prediction by the nonnegative squared term in equation (66). This is the curvature of the equilibrium relation. The transient comparison has to use the same initial level, inlet step and absolute/deviation convention in both models; otherwise a baseline mismatch can look like linearisation error.

Motivation and explanation

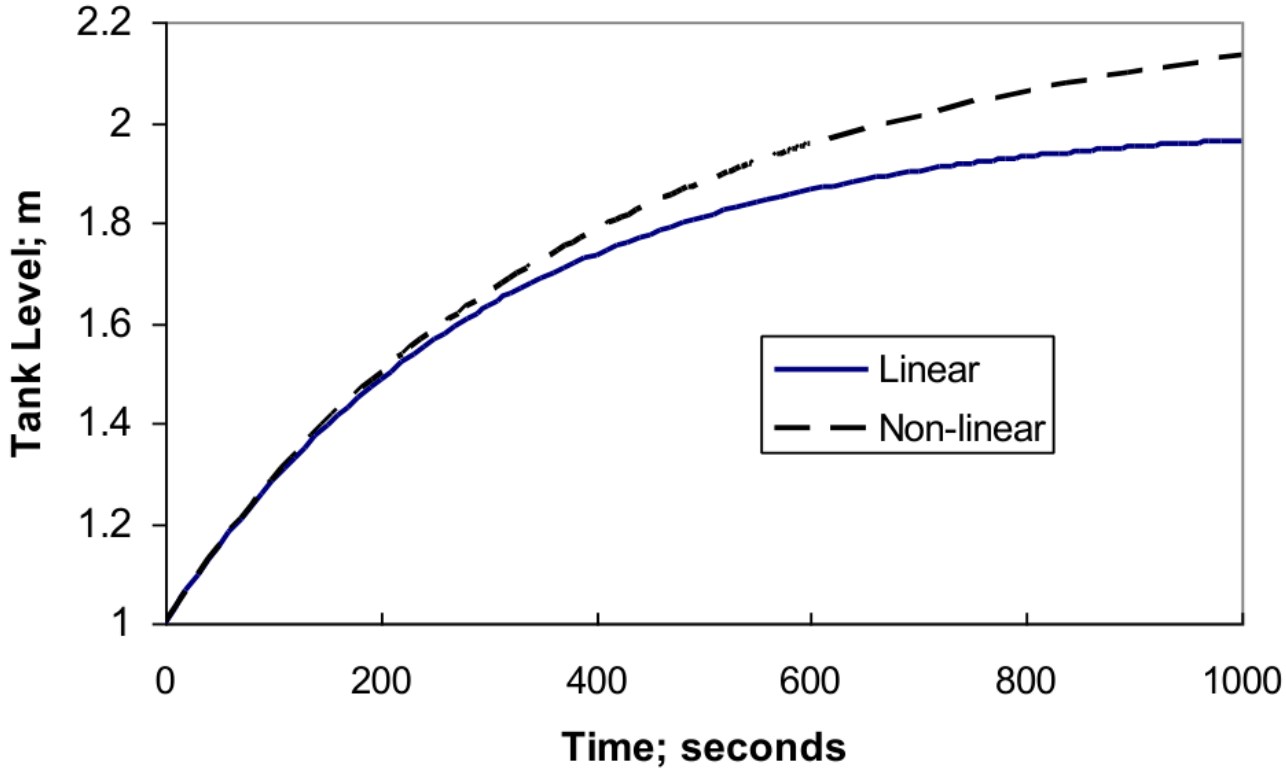
The tangent approximation needs $|h'|/h_s$ small. It fails near an empty tank, where the square-root derivative is singular. The coefficient k has units $\mathrm{m}^{5/2}\ \mathrm{s}^{-1}$, so both the original flow law and the derived time constant are dimensionally consistent.

The exact transient approaches equilibrium asymptotically. A numerical solution of the implicit equation or direct integration can quantify deviations from the local lag without inventing a constant

time constant for the nonlinear process.

Worked example

Take $A = 3.00$ m^2, $k = 0.0200$ m$^{5/2}$ s^{-1}, $q_{0s} = 0.0200$ m^3 s^{-1} and $\Delta q_0 = 0.0100$ m^3 s^{-1}. Then $h_s = 1.00$ m, $\tau = 300$ s and $K_p = 100$ s m^{-2}. The linear final level is 2.00 m; the exact final level is 2.25 m.



At 1000 s, the linear model gives $h = 1.96433$ m and direct nonlinear integration gives $h = 2.13198$ m. The implicit solution returns the same nonlinear value. The figure is qualitative; these values are computed from the balances, not read from the plotted lines. The large inlet change demonstrates why local linearisation must be checked for the intended excursion.

The graph illustrates two differences: the local model approaches a different endpoint and follows a different trajectory on the way there. Here the exact final rise exceeds the linear final rise because the true outflow increases less rapidly with level than its initial tangent predicts. To balance the larger feed, the exact model must reach a greater depth. A comparison of settling fractions should therefore use each model's own endpoint.

Link to other topics

Gravity outflow turns an otherwise integrating level into a self-regulating lag. Its steady gain is the derivative of the nonlinear equilibrium relation at the chosen point.

Conclusion

Linearisation preserves the local restoring physics. The exact equilibrium and implicit transient quantify what it loses for a large change.

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PDC-22: Three-tank mixing: valve linearisation and step response

Overview

Assumed: component balances, deviation variables, series lags and local linearisation.

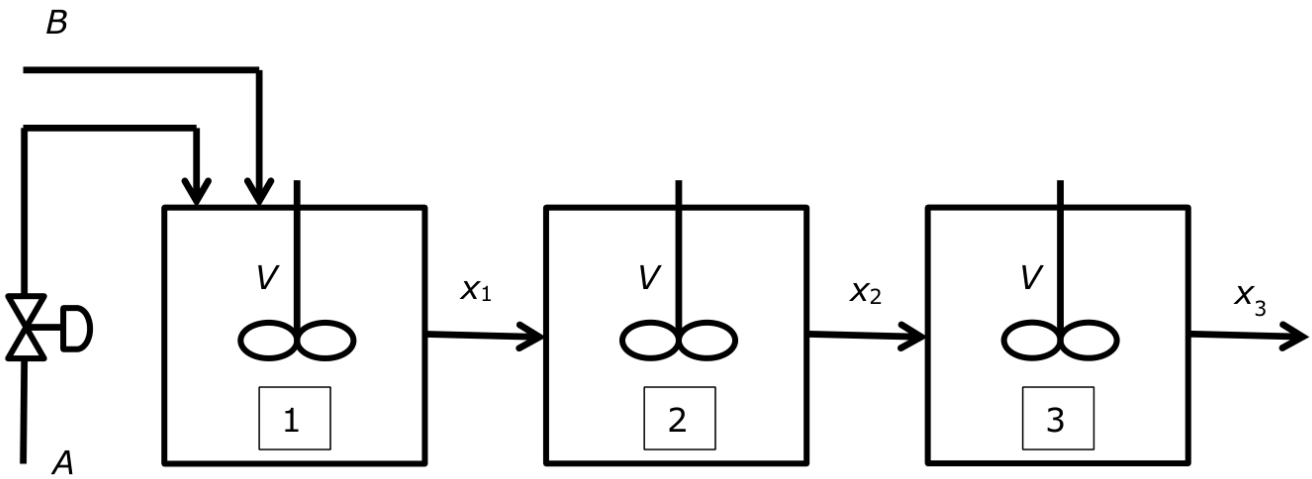
A three-tank mixer combines variable-flow nonlinearity with a cascade of concentration lags. The valve and disturbance gains must use a consistent steady composition and an explicit percentage convention.

Intuition

Opening the pure-tracer valve adds tracer, but it also increases total flow and changes residence time. The local model captures the first-order composition effect while dropping products of flow and concentration deviations. Downstream tanks smooth that change twice more.

Derivation

Step 1: define the composition convention and equilibrium. Three perfectly mixed tanks have equal fixed volume $V = 35$ m^3; their outlet flows follow the combined inlet flow, maintaining each volume. Stream A is pure inert tracer; stream B contains tracer concentration x_B expressed as a numerical **volume percent**, so pure A has $x_A = 100$, not 1. Assume equal constant densities and additive volumes.



Let flows be q_A, q_B in m^3 min^{-1}, tank volume-percent compositions x_1, x_2, x_3, and valve position v as a numerical percent open. A linear valve gives $q_A = k_v v$, $k_v = 0.0028$ m^3 min^{-1} per percentage point. At $v_s = 50$, $q_{As} = 0.14$; take $q_{Bs} = 6.86$ and $x_{Bs} = 2$. Then

$$x_s = \frac{100 q_{As} + q_{Bs} x_{Bs}}{q_{As} + q_{Bs}} = 3.96, \qquad \tau = \frac{V}{q_{As} + q_{Bs}} = 5.00 \text{ min}. \tag{68}$$

Reason: every tank has the same steady composition from the combined feed balance. Retain this equilibrium throughout the perturbation model.

Step 2: expand tank 1's balance. Hold $q_B = q_{Bs}$ while q_A and x_B can change:

$$V \dot{x}_1 = 100 q_A + q_{Bs} x_B - (q_A + q_{Bs}) x_1.$$

Substitute $q_A = q_{As} + q'_A$, $x_B = x_{Bs} + x'_B$ and $x_1 = x_s + x'_1$ and cancel the steady balance:

$$V \dot{x}'_1 = -(q_{As} + q_{Bs}) x'_1 + q_{Bs} x'_B + (100 - x_s) q'_A - q'_A x'_1.$$

Reason: the valve perturbation increases tracer inflow by $100 q'_A$ and removes more tank mixture by $x_s q'_A$; their difference sets its local concentration sensitivity. Neglect the product $q'_A x'_1$ and divide by total steady flow:

$$\tau \dot{x}'_1 + x'_1 = K_B x'_B + K_q q'_A, \quad K_B = \frac{q_{Bs}}{q_{As} + q_{Bs}}, \quad K_q = \frac{100 - x_s}{q_{As} + q_{Bs}}. \tag{69}$$

Step 3: linearise the downstream tanks. Tank 2 obeys $V \dot{x}_2 = (q_A + q_{Bs})(x_1 - x_2)$. Because its two baseline compositions are equal, the flow perturbation multiplies only a concentration deviation:

$$\tau \dot{x}'_2 + x'_2 = x'_1, \qquad \tau \dot{x}'_3 + x'_3 = x'_2, \qquad q'_A = k_v v'.$$

Reason: drop the second-order product of flow deviation and concentration difference. Transform and multiply the three lag factors:

$$\bar{x}_3 = \frac{K_B}{(\tau s + 1)^3} \bar{x}_B + \frac{K_v}{(\tau s + 1)^3} \bar{v}, \quad K_v = K_q k_v = 0.038416, \quad K_B = 0.980000. \tag{70}$$

Here s is in min^{-1}; K_v is concentration percentage points per valve-opening percentage point. The visible mode has pole $-1/5$ min^{-1} repeated three times.

Step 4: invert the valve step. Hold $x'_B = 0$ and apply $v' = \Delta v H(t)$. Write

$$\frac{1}{s(\tau s + 1)^3} = \frac{1}{s} - \frac{\tau}{\tau s + 1} - \frac{\tau}{(\tau s + 1)^2} - \frac{\tau}{(\tau s + 1)^3}.$$

Reason: multiplying by the common denominator telescopes to one. Inverting the repeated-pole terms gives

$$x'_3 = K_v \Delta v \left[1 - \left(1 + \frac{t}{\tau} + \frac{t^2}{2\tau^2} \right) e^{-t/\tau} \right]. \tag{71}$$

Reason: the third pole contributes the quadratic-in-time exponential term. With $\tau = 5$ min, its quadratic coefficient is $t^2/50$.

The first-tank valve gain contains $100 - x_s$ because each extra unit of pure-tracer inflow also drives out an extra unit of the existing mixture. Counting only the incoming tracer would overstate its effect. In a downstream tank, the baseline inlet and outlet compositions are identical, so changing flow alone creates no first-order composition forcing. It multiplies only the developing composition difference, which is why that contribution is a product of deviations.

Each tank adds one stored concentration and one lag factor. After a valve step, tank 1 begins changing first; its emerging change drives tank 2, which then drives tank 3. The polynomial multiplying the exponential in equation (71) enforces the initially gentle downstream response. A repeated pole describes several sequential memories with the same decay rate, not a single lag with its time constant multiplied by three.

Motivation and explanation

A volume-percent change from 2 to 3 is one **percentage point**; it is not a 1% relative increase. Using volume fractions is equally valid, but requires replacing 100 by 1 and rescaling concentration gains consistently.

The cascade has no exact dead time, though its initially flat response can resemble a delay. The local model freezes residence time at its baseline value and is accurate only for small flow/composition changes. A real valve may not be linear in stem position at changing pressure drop.

Worked example

A valve move from 50 to 60 percent open has $\Delta v = 10$ percentage points. The local model predicts a final tracer increase of 0.384160 percentage points. At 5.00 min the increase is 0.0308486 points; at 15.0 min it is 0.221587 points.

The final linear prediction is 4.34416%. Recomputing the exact steady feed mixture at $q_A = 0.168$ m^3 min^{-1} gives 4.34263%. Their small difference is due to the neglected flow–composition product. The distinction between the baseline 3.96% and a rounded 4% must not leave an uncancelled constant term in the derivation.

At one individual tank time constant the final outlet has completed only 8.03014% of its own final change; at three it has completed 57.6810%. The single-lag 63.2% rule therefore cannot be applied to the whole cascade using $\tau = 5$ min. The small difference between exact and linear final compositions checks the local endpoint approximation for this valve move; it does not by itself verify the whole transient.

Link to other topics

This example brings physical balances, local linearisation, transfer-function summation and repeated poles together. These plant-response models are the starting point for designing feedback controllers once a controller model is introduced.

Conclusion

A consistent equilibrium and percentage convention produce a clear local model. Its two input gains and three repeated lag factors separate mixture sensitivity from the dynamics of stored liquid.

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